

8.4 Sturm's Theorem

Consider a polynomial $f(x) \in K[x]$ where K is a real closed field. A classical technique due to Sturm shows how to compute the real zeros of $f(x)$ in an interval $[a, b]$. The recipe is as follows:

1. First compute a special sequence of polynomials $\overline{\text{STURM}}(f) = \langle h_0(x) = f(x), h_1(x), \dots, h_s(x) \rangle$, which will be called a *Sturm sequence*.
2. Next compute the “variations in sign” for the sequences $\overline{\text{STURM}}(f)(a) = \langle h_0(a), h_1(a), \dots, h_s(a) \rangle$ and $\overline{\text{STURM}}(f)(b) = \langle h_0(b), h_1(b), \dots, h_s(b) \rangle$ —denoted, respectively, by $\text{Var}_a(\overline{\text{STURM}}(f))$ and $\text{Var}_b(\overline{\text{STURM}}(f))$.
3. Then

$$\# \text{ real zeros of } f(x) \text{ in } (a, b) = \text{Var}_a(\overline{\text{STURM}}(f)) - \text{Var}_b(\overline{\text{STURM}}(f)).$$

However, these notions need to be formalized further. In this section, we shall start with a slightly general version of the statement above: the *Sturm-Tarski theorem*. Later on we shall study some generalizations to higher dimensions.

Definition 8.4.1 (Variations in Sign) If $\bar{c} = \langle c_1, \dots, c_m \rangle$ is a finite sequence of nonzero elements of a real closed field K , then we define the *number of variations in sign* of $\bar{c}$ to be

$$\left| \left\{ i : 1 \leq i < m \text{ and } c_i \cdot c_{i+1} < 0 \right\} \right|.$$

In general, If $\bar{c} = \langle c_1, \dots, c_m \rangle$ is an arbitrary sequence of elements of K , then we define the *number of variations in sign* of $\bar{c}$, denoted $\text{Var}(\bar{c})$, to be the number of variations in the sign of the abbreviated sequence $\text{abb}(\bar{c})$, which is obtained by omitting the zeros in $\bar{c}$.

Thus, $\text{Var}(\bar{c})$ is the number of times the entries of $\bar{c}$ change sign when scanned sequentially from left to right. $\square$

For example, $\langle 1, 0, 0, 2, -1, 0, 3, 4, -2 \rangle$ has three variations in sign. Note that for any nonzero $a \in K$

$$\text{Var}(\bar{c}) = \text{Var}(a\bar{c}),$$

where $a\langle c_1, \dots, c_m \rangle = \langle a c_1, \dots, a c_m \rangle$. Similarly, if $c_i \cdot c_{i+1} < 0$ then for any $a \in K$

$$\text{Var}(\langle c_1, \dots, c_i, c_{i+1}, \dots, c_m \rangle) = \text{Var}(\langle c_1, \dots, c_i, a, c_{i+1}, \dots, c_m \rangle).$$

For a vector of polynomials $\bar{F} = \langle f_1(x), \dots, f_m(x) \rangle \in K[x]^m$ and a field element $a \in K$, we write $\text{Var}_a(\bar{F})$ for $\text{Var}(\bar{F}(a))$:

$$\text{Var}_a(\bar{F}) = \text{Var}(\bar{F}(a)) = \text{Var}(\langle f_1(a), \dots, f_m(a) \rangle).$$

Definition 8.4.2 (Standard Sturm Sequence) The *standard Sturm sequence* (or, *canonical Sturm sequence*) of a pair of polynomials $f(x)$ and $g(x) \in K[x]$ ($K = \text{a field}$) is

$$\overline{\text{STURM}}(f, g) = \langle h_0(x), h_1(x), \dots, h_s(x) \rangle,$$

where

$$\begin{aligned} h_0(x) &= f(x) \\ h_1(x) &= g(x) \\ h_0(x) &= q_1(x) h_1(x) - h_2(x), & \deg(h_2) < \deg(h_1) \\ &\vdots \\ h_{i-1}(x) &= q_i(x) h_i(x) - h_{i+1}(x), & \deg(h_{i+1}) < \deg(h_i) \\ &\vdots \\ h_{s-1}(x) &= q_s(x) h_s(x). \quad \square \end{aligned}$$

Note that the standard sequence is termwise similar to the polynomial remainder sequence, except that we take the negation of the remainder at each step, i.e., it is based on the sequence $\langle 1, 1, -1, -1, \dots \rangle$. Also note that for any $f(x)$ and $g(x)$, the last element in their standard Sturm sequence, $\overline{\text{STURM}}(f, g)$, is in fact their GCD (up to a sign); in particular, $h_s(x) \mid h_i(x)$ ($0 \leq i \leq s$). In this case, we may consider their “*suppressed*” Sturm sequence,

$$\tilde{h}_i(x) = \frac{h_i(x)}{h_s(x)}.$$

Note, $\tilde{h}_s(x) = 1$. Furthermore, for any $i > 0$, we also have

$$\tilde{h}_{i-1}(x) = q_i(x) \tilde{h}_i(x) - \tilde{h}_{i+1}(x).$$

Lemma 8.4.1 Let $f(x)$ and $g(x) \in K[x]$, $K = \text{a real closed field}$, be a pair of polynomials with a standard Sturm sequence,

$$\overline{\text{STURM}}(f, g) = H = \langle h_0(x), h_1(x), \dots, h_s(x) \rangle.$$

Let $[a, b]$ be an interval ($a < b$) not containing a zero of $f(x) = h_0(x)$. Then

$$\text{Var}_a(\overline{\text{STURM}}(f, g)) - \text{Var}_b(\overline{\text{STURM}}(f, g)) = 0.$$

PROOF.

Let us consider an arbitrary subinterval $[a', b'] \subseteq [a, b]$ such that it contains at most one zero of the suppressed sequence

$$\tilde{H} = \langle \tilde{h}_0(x), \tilde{h}_1(x), \dots, \tilde{h}_s(x) \rangle.$$

It suffices to show that

$$\text{Var}_{a'}(\overline{\text{STURM}}(f, g)) - \text{Var}_{b'}(\overline{\text{STURM}}(f, g)) = 0.$$

Since the interval does not contain a zero of h_0 , nor does it contain a zero of $\tilde{h}_0$.

Now, if the interval does not contain a zero of any element of $\tilde{H}$, then $\tilde{h}_i(a') \tilde{h}_i(b') > 0$ for all i and

$$\text{Var}_{a'}(\tilde{H}) - \text{Var}_{b'}(\tilde{H}) = 0.$$

Alternatively, assume that the interval contains a zero of $\tilde{h}_i \in \tilde{H}$, i.e., for some $c \in [a', b']$, $\tilde{h}_i(c) = 0$ ($i > 0$). Then

$$\begin{aligned} \tilde{h}_{i-1}(c) &= q_i(c) \tilde{h}_i(c) - \tilde{h}_{i+1}(c) = -\tilde{h}_{i+1}(c) \\ \Rightarrow \tilde{h}_{i-1}(c) \tilde{h}_{i+1}(c) &= -\tilde{h}_{i+1}^2(c) < 0. \end{aligned}$$

Since $\tilde{h}_{i+1}(c) = 0$ would imply that $\tilde{h}_i(c) = \tilde{h}_{i+1}(c) = \dots = \tilde{h}_s(c) = 0$, contradicting the fact that $\tilde{h}_s = 1$. Thus,

$$\begin{aligned} \tilde{h}_{i-1}(c) \cdot \tilde{h}_{i+1}(c) &< 0 \\ \Rightarrow \tilde{h}_{i-1}(a') \cdot \tilde{h}_{i+1}(a') &< 0 \wedge \tilde{h}_{i-1}(b') \cdot \tilde{h}_{i+1}(b') < 0, \end{aligned}$$

which results in

$$\begin{aligned} \text{Var}_{a'}(\tilde{H}) &= \text{Var}(\tilde{h}_0(a'), \dots, \tilde{h}_{i-1}(a'), \tilde{h}_{i+1}(a'), \dots, \tilde{h}_s(a')) \\ &= \text{Var}(\tilde{h}_0(b'), \dots, \tilde{h}_{i-1}(b'), \tilde{h}_{i+1}(b'), \dots, \tilde{h}_s(b')) \\ &= \text{Var}_{b'}(\tilde{H}). \end{aligned}$$

Thus, in either case

$$\text{Var}_{a'}(\tilde{H}) - \text{Var}_{b'}(\tilde{H}) = 0.$$

Thus

$$\begin{aligned} &\text{Var}_{a'}(\overline{\text{STURM}}(f, g)) - \text{Var}_{b'}(\overline{\text{STURM}}(f, g)) \\ &= \text{Var}_{a'}(h_s(a') \tilde{H}) - \text{Var}_{b'}(h_s(b') \tilde{H}) \\ &= \text{Var}_{a'}(\tilde{H}) - \text{Var}_{b'}(\tilde{H}) \\ &= 0. \end{aligned}$$

Clearly, the interval $[a, b]$ can be partitioned into finitely many subintervals such that over each subinterval there is no net change in the variation, thus proving that there is no net change over the entire interval $[a, b]$, either. $\square$

Let $D: K[x] \rightarrow K[x]$ be the (formal) derivative map:

$$\begin{aligned} D &: x^n \mapsto n \cdot x^{n-1} \\ &: a \mapsto 0, \quad a \in K. \end{aligned}$$

Assume that $f(x) \in K[x]$ is an arbitrary polynomial over a real closed field K and $f'(x) = D(f(x))$ its *formal derivative* with respect to x .

Lemma 8.4.2 *Let $f(x)$ and $g(x) \in K[x]$, K = a real closed field. Consider a standard Sturm sequence of the polynomials $f(x)$ and $f'(x)g(x)$,*

$$\overline{\text{STURM}}(f, f'g) = H = \langle h_0(x), h_1(x), \dots, h_s(x) \rangle.$$

Let $[a, b]$ be an interval ($a < b$) containing exactly one zero $c \in (a, b)$ of $f(x) = h_0(x)$. Then

$$\text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)) = \text{sgn}(g(c)).$$

PROOF.

Without loss of generality, we may assume that none of the $h_i(x)$'s vanish in either of the half-open intervals: $[a, c)$ and $(c, b]$.

Let us write $f(x)$ and $g(x)$ as follows:

$$f(x) = (x - c)^r \phi(x) \quad \text{and} \quad g(x) = (x - c)^s \psi(x), \quad r > 0, \quad s \geq 0,$$

where, by assumption, we have $\phi(c) \neq 0$ and $\psi(c) \neq 0$.

Then,

$$f'(x) = (x - c)^{r-1} [r\phi(x) + (x - c)\phi'(x)].$$

Thus

$$f(x) f'(x) g(x) = (x - c)^{2r+s-1} [r\phi^2(x)\psi(x) + (x - c)\phi(x)\phi'(x)\psi(x)].$$

Now, we are ready to consider each of the cases:

- CASE 1: $s = 0$, i.e., $g(c) \neq 0$.

In that case,

$$f(x) f'(x) g(x) = (x - c)^{2r-1} [r\phi^2(x)\psi(x) + (x - c)\phi(x)\phi'(x)\psi(x)],$$

an *odd function* of x in the neighborhood of $c \in [a, b]$. If $g(c) > 0$, then in the neighborhood of c ,

$$f(x) f'(x) g(x) = (x - c)^{2r-1} [k_+ + \epsilon],$$

where $k_+ = r\phi^2(c)\psi(c) > 0$. Thus to the left of c , $f(x)$ and $f'(x)g(x)$ have opposite signs and to the right same signs, implying a *loss of sign* as one moves from left to right past c :

$$\text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)) = +1.$$

Similarly, if $g(c) < 0$, then in the neighborhood of c ,

$$f(x) f'(x) g(x) = (x - c)^{2r-1} [k_- + \epsilon],$$

where $k_- = r\phi^2(c)\psi(c) < 0$. Thus to the left of c $f(x)$ and $f'(x)g(x)$ have same signs and to the right opposite signs, implying a *gain of sign*:

$$\text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)) = -1.$$

- CASE 2: $s > 0$, i.e., $g(c) = 0$.

1. SUBCASE 2A: $s = 1$.

In this case

$$\begin{aligned} h_0(x) = f(x) &= (x - c)^r \phi(x), \quad \text{and} \\ h_1(x) = f'(x)g(x) &= (x - c)^r [r\phi(x)\psi(x) + (x - c)\phi'(x)\psi(x)]. \end{aligned}$$

Thus

$$\begin{aligned} \tilde{h}_0(x) &= \frac{h_0(x)}{(x - c)^r} = \phi(x), \\ \tilde{h}_1(x) &= \frac{h_1(x)}{(x - c)^r} = r\phi(x)\psi(x) + (x - c)\phi'(x)\psi(x). \end{aligned}$$

Hence the suppressed sequence has no zero in the interval $[a, b]$, and thus the suppressed sequence undergoes no net variation of signs. Arguing as in the previous lemma, we have:

$$\text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)) = 0.$$

2. SUBCASE 2B: $s > 1$.

In this case $\deg(h_0) = \deg(f) < \deg(f'g) = \deg(h_1)$; thus,

$$h_2(x) = -h_0(x),$$

i.e., the first and third entry in the sequence have exactly the opposite signs. Again considering the suppressed sequence, we see that the suppressed sequence (and hence the original sequence) suffers zero net variation of signs. Hence:

$$\text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)) = 0. \quad \square$$

Theorem 8.4.3 (General Sturm-Tarski Theorem) *Let $f(x)$ and $g(x)$ be two polynomials with coefficients in a real closed field K and let*

$$\overline{\text{STURM}}(f, f'g) = \left\langle \begin{array}{l} h_0(x) = f(x), \\ h_1(x) = f'(x)g(x), \\ h_2(x), \\ \vdots \\ h_s(x) \end{array} \right\rangle,$$

where h_i 's are related by the following relations ($i > 0$):

$$h_{i-1}(x) = q_i(x) h_i(x) - h_{i+1}(x), \quad \deg(h_{i+1}) < \deg(h_i).$$

Then for any interval $[a, b] \subseteq K$ ($a < b$):

$$\text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b = c_f [g > 0]_a^b - c_f [g < 0]_a^b,$$

where

$$\text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b \triangleq \text{Var}_a(\overline{\text{STURM}}(f, f'g)) - \text{Var}_b(\overline{\text{STURM}}(f, f'g)).$$

and $c_f[P]_a^b$ counts the number of distinct roots ($\in K$, and without counting multiplicity) of f in the interval $(a, b) \subseteq K$ at which the predicate P holds.

PROOF.

Take all the roots of all the polynomials $h_j(x)$'s in the Sturm sequence, and decompose the interval $[a, b]$ into finitely many subintervals each containing at most one of these roots.

The rest follows from the preceding two lemmas, since

$$\text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b = \sum_{c \in (a, b), f(c)=0} \text{sgn}(g(c)). \quad \square$$

Corollary 8.4.4 *Let $f(x)$ be a polynomial with coefficients in a real closed field K .*

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0.$$

Then

1. For any interval $[a, b] \subseteq K$ ($a < b$):

$$\begin{aligned} \text{Var} \left[\overline{\text{STURM}}(f, f') \right]_a^b \\ = \# \text{distinct roots } \in K \text{ of } f \text{ in the interval } (a, b). \end{aligned}$$

2. Let $L \in K$ be such that all the roots of $f(x)$ are in the interval $(-L, +L)$; e.g.,

$$L = \frac{|a_n| + \max(|a_{n-1}|, \dots, |a_0|)}{|a_n|}.$$

Then the total number of distinct roots of f in K is given by

$$\text{Var} \left[\overline{\text{STURM}}(f, f') \right]_{-L}^{+L} = \text{Var}_{-L}(\overline{\text{STURM}}(f, f')) - \text{Var}_{+L}(\overline{\text{STURM}}(f, f')).$$

PROOF.

(1) The first part is a corollary of the preceding theorem, with g taken to be the constant positive function 1.

(2) The second part follows from the first, once we observe that all the roots of f lie in the interval $[-L, +L]$. $\square$

Corollary 8.4.5 *Let $f(x)$ and $g(x)$ be two polynomials with coefficients in a real closed field K , and assume that*

$$f(x) \text{ and } f'(x)g(x) \text{ are relatively prime.}$$

Then

$$\text{Var} \left[\overline{\text{STURM}}(f, g) \right]_a^b = c_f \left[(f'g) > 0 \right]_a^b - c_f \left[(f'g) < 0 \right]_a^b. \quad \square$$

Corollary 8.4.6 *Let $f(x)$ and $g(x)$ be two polynomials with coefficients in a real closed field K . For any interval $[a, b] \subseteq K$ ($a < b$), we have*

$$\begin{aligned} c_f \left[g = 0 \right]_a^b + c_f \left[g > 0 \right]_a^b + c_f \left[g < 0 \right]_a^b &= \text{Var} \left[\overline{\text{STURM}}(f, f') \right]_a^b, \\ c_f \left[g > 0 \right]_a^b - c_f \left[g < 0 \right]_a^b &= \text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b, \\ c_f \left[g > 0 \right]_a^b + c_f \left[g < 0 \right]_a^b &= \text{Var} \left[\overline{\text{STURM}}(f, f'g^2) \right]_a^b. \quad \square \end{aligned}$$

Thus the previous corollary can be expressed as follows,

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_f \left[g = 0 \right]_a^b \\ c_f \left[g > 0 \right]_a^b \\ c_f \left[g < 0 \right]_a^b \end{bmatrix} = \begin{bmatrix} \text{Var} \left[\overline{\text{STURM}}(f, f') \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g^2) \right]_a^b \end{bmatrix},$$

or equivalently:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \text{Var} \left[\overline{\text{STURM}}(f, f') \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g) \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g^2) \right]_a^b \end{bmatrix} = \begin{bmatrix} c_f \left[g = 0 \right]_a^b \\ c_f \left[g > 0 \right]_a^b \\ c_f \left[g < 0 \right]_a^b \end{bmatrix}.$$

8.5 Real Algebraic Numbers

In this section, we study how real algebraic numbers may be described and manipulated. We shall introduce some machinery for this purpose, i.e., root separation and Thom's lemma.

8.5.1 Real Algebraic Number Field

Consider a field E . Let F be subfield in E . An element $u \in E$ is said to be *algebraic* over F if for some nonzero polynomial $f(x) \in F[x]$, $f(u) = 0$; otherwise, it is *transcendental* over F .

Similarly, let S be a subring of E . An element $u \in E$ is said to be *integral* over S if for some monic polynomial $f(x) \in S[x]$, $f(u) = 0$.

For example, if we take $E = \mathbb{C}$, and $F = \mathbb{Q}$, then the elements of $\mathbb{C}$ that are algebraic over $\mathbb{Q}$ are the *algebraic numbers*; they are simply the algebraic closure of $\mathbb{Q}$: $\overline{\mathbb{Q}}$. Similarly, if we take $S = \mathbb{Z}$, then the elements of $\mathbb{C}$ that are integral over $\mathbb{Z}$ are the *algebraic integers*.

Other useful examples are obtained by taking $E = \mathbb{R}$, $F = \mathbb{Q}$ and $S = \mathbb{Z}$; they give rise to the *real algebraic numbers* and *real algebraic integers*—topics of this section; they are in a very real sense a significant fragment of “computable numbers” and thus very important.

Definition 8.5.1 (Real Algebraic Number) A real number is said to be a *real algebraic number* if it is a root of a univariate polynomial $f(x) \in \mathbb{Z}[\cdot]$ with integer coefficients. $\square$

Example 8.5.2 Some examples of real algebraic numbers:

1. All integers:

$$n \in \mathbb{Z} \text{ is a root of } x - n = 0.$$

2. All rational numbers:

$$\alpha = \frac{p}{q} \in \mathbb{Q} \ (q \neq 0) \text{ is a root of } qx - p = 0.$$

3. All real radicals of rational numbers:

$$\beta = \sqrt[n]{p/q} \in \mathbb{R} \ (p \geq 0, q > 0) \text{ is a root of } qx^n - p = 0. \quad \square$$

Definition 8.5.3 (Real Algebraic Integer) A real number is said to be a *real algebraic integer* if it is a root of a univariate monic polynomial $f(x) \in \mathbb{Z}[\cdot]$ with integer coefficients. $\square$

Example 8.5.4 The golden ratio,

$$\xi = \frac{1 + \sqrt{5}}{2},$$

is an algebraic integer, since ξ is a root of the monic polynomial

$$x^2 - x - 1. \quad \square$$

Lemma 8.5.1 *Every real algebraic number can be expressed as a real algebraic integer divided by an integer.*

(This is a corollary of the following general theorem; the proof given here is the same as the general proof, *mutatis mutandis*.
Every algebraic number is a ratio of an algebraic integer and an integer.)

PROOF.

Consider a real algebraic number ξ . By definition,

$$\begin{aligned} \xi &= \text{a real algebraic number} \\ \Leftrightarrow \xi &= \text{real root of a polynomial} \\ f(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[\curvearrowright], \quad \mathfrak{D}_\times \neq \mathcal{K}. \end{aligned}$$

Now, if we multiply $f(x)$ by a_n^{n-1} , we have

$$\begin{aligned} a_n^{n-1} f(x) &= a_n^{n-1} a_n x^n + a_n^{n-1} a_{n-1} x^{n-1} + \cdots + a_n^{n-1} a_0 \\ &= (a_n x)^n + a_{n-1} (a_n x)^{n-1} + \cdots + a_0 a_n^{n-1}. \end{aligned}$$

Clearly, $a_n \xi$ is a real root of the polynomial

$$g(y) = y^n + a_{n-1} y^{n-1} + \cdots + a_0 a_n^{n-1} \in \mathbb{Z}[\curvearrowright],$$

as $g(a_n \xi) = a_n^{n-1} f(\xi) = 0$. Thus $a_n \xi$ a real algebraic integer, for some $a_n \in \mathbb{Z} \setminus \{\mathcal{K}\}$. $\square$

Lemma 8.5.2

1. If α, β are real algebraic numbers, then so are $-\alpha, \alpha^{-1}$ ($\alpha \neq 0$), $\alpha + \beta$, and $\alpha \cdot \beta$.
2. If α, β are real algebraic integers, then so are $-\alpha, \alpha + \beta$, and $\alpha \cdot \beta$.

PROOF.

1. (a) If α is a real algebraic number (defined as a real root of $f(x) \in \mathbb{Z}[\curvearrowright]$), then $-\alpha$ is also a real algebraic number.

$$\begin{aligned} \alpha &= \text{a real root of } f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \\ \Leftrightarrow -\alpha &= \text{a real root of } a_n (-x)^n + a_{n-1} (-x)^{n-1} + \cdots + a_0 \\ \Leftrightarrow -\alpha &= \text{a real root of} \\ &\quad (-1)^n a_n x^n + (-1)^{n-1} a_{n-1} (-x)^{n-1} + \cdots + a_0 \in \mathbb{Z}[\curvearrowright]. \end{aligned}$$

- (b) If α is a real algebraic integer, then $-\alpha$ is also a real algebraic integer.

2. If α is a nonzero real algebraic number (defined as a real root of $f(x) \in \mathbb{Z}[\curvearrowright]$), then $1/\alpha$ is also a real algebraic number.

$$\alpha = \text{a real root of } f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$$

$$\Leftrightarrow 1/\alpha = \text{a real root of}$$

$$x^n a_n \left(\frac{1}{x}\right)^n + x^n a_{n-1} \left(\frac{1}{x}\right)^{n-1} + \cdots + x^n a_0$$

$$\Leftrightarrow 1/\alpha = \text{a real root of } a_0 x^n + a_1 x^{n-1} + \cdots + a_n \in \mathbb{Z}[\curvearrowright].$$

Note that, if α is a nonzero real algebraic integer, then $1/\alpha$ is a real algebraic number, but not necessarily a real algebraic integer.

3. (a) If α and β are real algebraic numbers (defined as real roots of $f(x)$ and $g(x) \in \mathbb{Z}[\curvearrowright]$, respectively), then $\alpha + \beta$ is also a real algebraic number, defined as a real root of

$$\text{Resultant}_y(f(x-y), g(y)).$$

$$\alpha = \text{a real root of } f(x) \text{ and } \beta = \text{a real root of } g(x)$$

$$\Leftrightarrow x - \alpha = \text{a real root of } f(x-y) \in (\mathbb{Z}[\curvearrowright])[\curvearrowright],$$

$$\beta = \text{a real root of } g(y) \in (\mathbb{Z}[\curvearrowright])[\curvearrowright] \text{ and}$$

$$x - \alpha = \beta = \text{a common real root of } f(x-y) \text{ and } g(y)$$

$$\Leftrightarrow x = \alpha + \beta = \text{a real root of}$$

$$\text{Resultant}_y(f(x-y), g(y)) \in \mathbb{Z}[\curvearrowright].$$

- (b) If α and β are real algebraic integers, then $\alpha + \beta$ is also a real algebraic integer.

4. (a) If α and β are real algebraic numbers (defined as real roots of $f(x)$ and $g(x) \in \mathbb{Z}[\curvearrowright]$, respectively), then $\alpha\beta$ is also a real algebraic number, defined as a real root of

$$\text{Resultant}_y\left(y^m f\left(\frac{x}{y}\right), g(y)\right), \quad \text{where } m = \deg(f).$$

$$\alpha = \text{a real root of } f(x) \text{ and } \beta = \text{a real root of } g(x)$$

$$\Leftrightarrow \frac{x}{\alpha} = \text{a real root of } y^m f\left(\frac{x}{y}\right) \in (\mathbb{Z}[\curvearrowright])[\curvearrowright],$$

$$\beta = \text{a real root of } g(y) \in (\mathbb{Z}[\curvearrowright])[\curvearrowright] \text{ and}$$

$$\frac{x}{\alpha} = \beta = \text{a common real root of } y^m f\left(\frac{x}{y}\right) \text{ and } g(y)$$

$$\Leftrightarrow x = \alpha\beta = \text{a real root of}$$

$$\text{Resultant}_y\left(y^m f\left(\frac{x}{y}\right), g(y)\right) \in \mathbb{Z}[\curvearrowright].$$

- (b) If α and β are real algebraic integers, then $\alpha\beta$ is also a real algebraic integer. $\square$

Corollary 8.5.3

1. The real algebraic integers form a ring.
2. The real algebraic numbers form a field, denoted by $\mathbb{A}$. $\square$

Since an algebraic number α , by definition, is a root of a nonzero polynomial $f(x)$ over $\mathbb{Z}$, we may say that f is α 's *polynomial*. Additionally, if f is of minimal degree among all such polynomials, then we say that it is α 's *minimal polynomial*. The *degree* of a nonzero algebraic number is the degree of its minimal polynomial; and by convention, the *degree* of 0 is $-\infty$. It is not hard to see that an algebraic number has a unique minimal polynomial modulo *associativity*; that is, if $f(x)$ and $g(x)$ are two minimal polynomials of an algebraic number α , then $f(x) \approx g(x)$.

If we further assume that α is a real algebraic number, then we can talk about its *minimal polynomial* and *degree* just as before.

Theorem 8.5.4 *The field of real algebraic numbers, $\mathbb{A}$, is an Archimedean real closed field.*

PROOF.

Since $\mathbb{A} \subset \mathbb{R}$ and since $\mathbb{R}$ itself is an ordered field, the induced ordering on $\mathbb{A}$ defines the unique ordering.

$\mathbb{A}$ is Archimedean: Consider a real algebraic number α , defined by its minimal polynomial $f(x) \in \mathbb{Z}[\wedge]$:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

and let $N = 1 + \max(|a_{n-1}|, \dots, |a_0|) \in \mathbb{Z}$. Then $\alpha < N$.

$\mathbb{A}$ is real closed: Clearly every positive real algebraic number α (defined by its minimal polynomial $f(x) \in \mathbb{Z}[\wedge]$) has a square root $\sqrt{\alpha} \in \mathbb{A}$ defined by a polynomial $f(x^2) \in \mathbb{Z}[\wedge]$. Also if $f(x) \in \mathbb{A}[\wedge]$ is a polynomial of odd degree, then as its complex roots appear in pair, it must have at least one real root; it is clear that this root is in $\mathbb{A}$. $\square$

8.5.2 Root Separation, Thom's Lemma and Representation

Given a real algebraic number α , we will see that it can be finitely represented by its *polynomial* and some additional information that identifies the root, if α 's polynomial has more than one real root.

If we want a succinct representation, then we must represent α by its *minimal polynomial* or simply a polynomial of sufficiently small degree (e.g., by asking that its polynomial is square-free). In many cases, if we require that even the intermediate computations be performed with succinct

representations, then the cost of the computation may become prohibitive, as we will need to perform polynomial factorization over $\mathbb{Z}$ at each step. Thus the prudent choice seems to be to represent the inputs and outputs succinctly, while adopting a more flexible representation for intermediate computation.

Now coming back to the component of the representation that identifies the root, we have essentially three choices: *order* (where we assume the real roots are indexed from left to right), *sign* (by a vector of signs) and *interval* (an interval $[a, b] \subset \mathbb{R}$ that contains exactly one root). Again the choice may be predicated by the succinctness, the model of computation, and the application.

Before we go into the details, we shall discuss some of the necessary technical background: namely, *root separation*, *Fourier sequence* and *Thom's lemma*.

Root Separation

In this section, we shall study the distribution of the real roots of an integral polynomial $f(x) \in \mathbb{Z}[\wedge]$. In particular, we need to determine how small the distance between a pair of distinct real roots may be as some function of the *size* of their polynomial. Using these bounds, we will be able to construct an interval $[a, b]$ ($a, b \in \mathbb{Q}$) containing exactly one real root of an integral polynomial $f(x)$, i.e., an interval that can *isolate* a real root of $f(x)$.

We keep our treatment general by taking $f(x)$ to be an arbitrary polynomial (not just square-free polynomials). Other bounds in the literature include cases (1) where $f(x)$ may be a rational complex polynomial or a Gaussian polynomial; (2) where $f(x)$ is square-free or irreducible; or (3) when we consider complex roots of $f(x)$. For our purpose, it is sufficient to deal with the separation among the real roots of an integral polynomial.

Definition 8.5.5 (Real Root Separation) If the distinct real roots of $f(x) \in \mathbb{Z}[\wedge]$ are $\alpha_1, \dots, \alpha_l$ ($l \geq 2$),

$$\alpha_1 < \alpha_2 < \dots < \alpha_l,$$

then define the *separation*¹ of f to be

$$\text{Separation}(f) = \min_{1 \leq i < j \leq l} (\alpha_j - \alpha_i).$$

If f has less than two distinct roots, then $\text{Separation}(f) = \infty$. $\square$

The following bound is due to S.M. Rump:

¹Or more exactly, *real separation*.

Theorem 8.5.5 (Rump's Bound) *Let $f(x) \in \mathbb{Z}[\sqrt{}]$ be an integral polynomial as follows:*

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0, \quad a_i \in \mathbb{Z}.$$

Then²

$$\text{Separation}(f) > \frac{1}{n^{n+1}(1 + \|f\|_1)^{2n}}.$$

PROOF.

Let $h \in \mathbb{R}$ be such that, for any arbitrary real root, α , of f , the polynomial f remains nonzero through out the interval $(\alpha, \alpha + h)$. Then, clearly,

$$\text{Separation}(f) > h.$$

Using the intermediate value theorem (see Problem 8.2), we have

$$f(\alpha + h) = f(\alpha) + hf'(\mu), \quad \text{for some } \mu \in (\alpha, \alpha + h).$$

Since $f(\alpha) = 0$, we have

$$h = \frac{|f(\alpha + h)|}{|f'(\mu)|}.$$

Thus we can obtain our bounds by choosing h such that $|f(\alpha + h)|$ is “fairly large” and $|f'(\mu)|$, “fairly small,” for all $\mu \in (\alpha, \alpha + h)$. Thus the polynomial needs enough space to go from a “fairly large” value to 0 at a “fairly small” rate.

Let β be a real root of $f(x)$ such that β is immediately to the right of α . Then there are following two cases to consider:

- CASE 1: $|\alpha| < 1 < |\beta|$.

We consider the situation $-1 < \alpha < 1 < \beta$, as we can always achieve this, by replacing $f(x)$ by $f(-x)$, if necessary. Take $\alpha + h = 1$. Then

1. $|f(\alpha + h)| \geq 1$, since $f(1) \neq 0$.
2. Since $\mu \leq 1$, we have

$$|f'(\mu)| \leq |na_n| + |(n-1)a_{n-1}| + \cdots + |a_1| \leq n\|f\|_1.$$

Thus

$$\text{Separation}(f) > h \geq \frac{1}{n\|f\|_1} > \frac{1}{n^{n+1}(1 + \|f\|_1)^{2n}}.$$

²Note: A tighter bound is also given by Rump:

$$\text{Separation}(f) > \frac{2\sqrt{2}}{n^{n/2+1}(1 + \|f\|_1)^n}.$$

For the next case, we need the following technical lemma:

Lemma 8.5.6 *Let $f(x) \in \mathbb{Z}[\sqrt{\cdot}]$ be an integral polynomial as in the theorem. If γ satisfies*

$$f'(\gamma) = 0, \quad \text{but} \quad f(\gamma) \neq 0,$$

then

$$|f(\gamma)| > \frac{1}{n^n(1 + \|f\|_1)^{2n-1}}.$$

PROOF.

Consider the polynomials $f'(x)$ and $\tilde{f}(x, y) = f(x) - y$. Since γ is a zero of $f'(x)$ and $\langle \gamma, f(\gamma) \rangle$ is a zero of $\tilde{f}(x, y)$, we see that $f(\gamma) \neq 0$ is a root of

$$R(y) = \text{Resultant}_x(f'(x), \tilde{f}(x, y)).$$

Using the Hadamard-Collins-Horowitz inequality theorem (with 1-norm) (see Problem 8.11), we get

$$\|R\|_\infty \leq (\|f\|_1 + 1)^{n-1} (n\|f\|_1)^n,$$

and

$$1 + \|R\|_\infty < n^n(1 + \|f\|_1)^{2n-1}.$$

Since $f(\gamma)$ is a nonzero root of the integral polynomial $R(y)$, we have

$$|f(\gamma)| > \frac{1}{1 + \|R\|_\infty} > \frac{1}{n^n(1 + \|f\|_1)^{2n-1}}. \quad \square$$

(End of Lemma)

- CASE 2: $|\alpha|, |\beta| \leq 1$.

By Rolle's theorem there is a $\gamma \in (\alpha, \beta)$, where $f'(\gamma) = 0$. Take $\alpha + h = \gamma$. Then

1. By Lemma 8.5.6,

$$|f(\alpha + h)| = |f(\gamma)| > \frac{1}{n^n(1 + \|f\|_1)^{2n-1}}.$$

2. As before, since $\mu \leq 1$, we have

$$|f'(\mu)| \leq n\|f\|_1.$$

Thus

$$\begin{aligned} \text{Separation}(f) &> h \\ &> \frac{1}{n^n(1 + \|f\|_1)^{2n-1}(n\|f\|_1)} \\ &> \frac{1}{n^{n+1}(1 + \|f\|_1)^{2n}}. \end{aligned}$$

Note: The remaining case $|\alpha|, |\beta| \geq 1$, requires no further justification as the following argument shows: Consider the polynomial $x^n f(1/x)$. Then α^{-1} and β^{-1} are two consecutive roots of the new polynomial of same “size” as $f(x)$; $|\alpha^{-1}|, |\beta^{-1}| \leq 1$; and

$$\beta - \alpha \geq \frac{\beta - \alpha}{\alpha \beta} = \alpha^{-1} - \beta^{-1}.$$

But using case 2, we get

$$\text{Separation}(f) > \beta - \alpha > \frac{1}{n^{n+1}(1 + \|f\|_1)^{2n}}. \quad \square$$

Let $r = p/q \in \mathbb{Q}$ be a rational number ($p, q \in \mathbb{Z}$). We define *size* of r as

$$\text{size}(r) = |p| + |q|.$$

Theorem 8.5.7 *Let $f(x) \in \mathbb{Z}[\wedge]$ be an integral polynomial of degree n . Then between any two real roots of $f(x)$ there is a rational number r of size*

$$\text{size}(r) < 2 \cdot n^{n+1}(1 + \|f\|_1)^{2n+1}.$$

PROOF.

Consider two consecutive roots of $f(x)$: $\alpha < \beta$. Let $Q \in \mathbb{Z}$ be

$$Q = n^{n+1}(1 + \|f\|_1)^{2n}.$$

Consider all the rational numbers with Q in the denominator:

$$\dots, -\frac{i}{Q}, -\frac{i-1}{Q}, \dots, -\frac{2}{Q}, -\frac{1}{Q}, 0, \frac{1}{Q}, \frac{2}{Q}, \dots, \frac{i-1}{Q}, \frac{i}{Q}, \dots$$

Since

$$\beta - \alpha > 1/Q,$$

for some P ,

$$\alpha < \frac{P}{Q} < \beta < \|f\|_1,$$

since β is a root of f .

Hence

$$|P| < n^{n+1}(1 + \|f\|_1)^{2n+1}.$$

Thus the size of $r = P/Q$ is

$$\text{size}(r) < 2 \cdot n^{n+1}(1 + \|f\|_1)^{2n+1}. \quad \square$$

Definition 8.5.6 (Isolating Interval) An *isolating interval* for an integral polynomial $f(x) \in \mathbb{Z}[\wedge]$ is an interval $[a, b]$ ($a, b \in \mathbb{Q}$), in which there is precisely one real root of the polynomial. $\square$

By the preceding theorem, we see that for every real root of an integral polynomial, we can find an isolating interval. Furthermore,

Corollary 8.5.8 *Every isolating interval of a degree n integral polynomial $f(x) \in \mathbb{Z}[\curvearrowright]$ is of the form $[a, b]$, ($a, b \in \mathbb{Q}$) and $\text{size}(a)$ and $\text{size}(b)$ are no larger than*

$$2 \cdot n^{n+1} (1 + \|f\|_1)^{2n+1}. \quad \square$$

Hence every isolating interval can be represented with $O(n(\lg n + \beta(f)))$ bits, where $\beta(f) = O(\lg \|f\|_1)$ is the bit-complexity of the polynomial $f(x)$ in any reasonable binary encoding. An important related computational problem is the so-called *root isolation problem*, where we are asked to compute an isolating interval of an integral polynomial $f(x)$.

ROOTISOLATION($f(x)$)

Input: $f(x) \in \mathbb{Z}[\curvearrowright]$; $\deg(f) = n$.

Output: An isolating interval $[a, b]$ ($a, b \in \mathbb{Q}$),
containing a real root of $f(x)$.

Let $\overline{S} = \overline{\text{STURM}}(f, f')$, be a Sturm sequence;
 $a := -\|f\|_1$; $b := \|f\|_1$;

if $\text{Var}_a(\overline{S}) = \text{Var}_b(\overline{S})$ then return failure;

while $\text{Var}_a(\overline{S}) - \text{Var}_b(\overline{S}) > 1$ loop
 $c := (a + b)/2$;
 if $\text{Var}_a(\overline{S}) > \text{Var}_c(\overline{S})$ then $b := c$ else $a := c$
 end{if };
end{loop };

return $[a, b]$;
end{ROOTISOLATION} $\square$

Note that each polynomial of the Sturm sequence can be evaluated by $O(n)$ arithmetic operations, and since there are at most n polynomials in the sequence, each successive refinement of the interval $[a, b]$ (by binary division) takes $O(n^2)$ time. Thus the time complexity of root isolation is

$$O(n^2) \cdot O\left(\lg\left(\frac{2\|f\|_1}{\text{Separation}(f)}\right)\right),$$

which is simplified to

$$\begin{aligned} & O\left(n^2 \lg\left(\frac{2\|f\|_1}{n^{-n-1}(1 + \|f\|_1)^{-2n}}\right)\right) \\ &= O\left(n^2 \lg(2n^{n+1}(1 + \|f\|_1)^{2n+1})\right) \\ &= O(n^3(\lg n + \beta(f))), \end{aligned}$$

where $\beta(f)$, as before, is the bit-complexity of the polynomial $f(x)$.

Fourier Sequence and Thom's Lemma

Definition 8.5.7 (Fourier Sequence) Let $f(x) \in \mathbb{R}[\curvearrowright]$ be a real univariate polynomial of degree n . Its *Fourier sequence* is defined to be the following sequence of polynomials:

$$\overline{\text{FOURIER}}(f) = \left\langle \begin{array}{l} f^{(0)}(x) = f(x), \\ f^{(1)}(x) = f'(x), \\ f^{(2)}(x), \\ \vdots \\ f^{(n)}(x) \end{array} \right\rangle,$$

where $f^{(i)}$ denotes the i^{th} derivative of f with respect to x . $\square$

Note that $\overline{\text{FOURIER}}(f')$ is a suffix of $\overline{\text{FOURIER}}(f)$ of length n ; in general, $\overline{\text{FOURIER}}(f^{(i)})$ is a suffix of $\overline{\text{FOURIER}}(f)$ of length $n - i + 1$.

Lemma 8.5.9 (Little Thom's Lemma) Let $f(x) \in \mathbb{R}[\curvearrowright]$ be a real univariate polynomial of degree n . Given a sign sequence $\overline{s}$:

$$\overline{s} = \langle s_0, s_1, s_2, \dots, s_n \rangle,$$

we define the sign-invariant region of $\mathbb{R}$ determined by $\overline{s}$ with respect to $\overline{\text{FOURIER}}(f)$ as follows:

$$R(\overline{s}) = \left\{ \xi \in \mathbb{R} : \text{sgn}(\mathcal{U}^{(\beth)}(\xi)) = \sim \beth, \text{ for all } \beth = \mathcal{V}, \dots, \mathcal{X} \right\}.$$

Then every nonempty $R(\overline{s})$ must be connected, i.e., consist of a single interval.

PROOF.

The proof is by induction on n . The base case (when $n = 0$) is trivial, since, in this case, either $R(\overline{s}) = \mathbb{R}$ (if $\text{sgn}(f(x)) = s_0$) or $R(\overline{s}) = \emptyset$ (if $\text{sgn}(f(x)) \neq s_0$).

Consider the induction case when $n > 0$. By the inductive hypothesis, we know that the sign-invariant region determined by $\overline{s}'$,

$$\overline{s}' = \langle s_1, s_2, \dots, s_n \rangle,$$

with respect to $\overline{\text{FOURIER}}(f')$ is either $\emptyset$ or a single interval. If it is empty, then there is nothing to prove. Thus we may assume that

$$R(\overline{s}') \neq \emptyset.$$

Now, let us enumerate the distinct real roots of $f(x)$:

$$\xi_1 < \xi_2 < \cdots < \xi_m.$$

Note that if $R(\overline{s}')$ consists of more than one interval, then a subsequence of real roots

$$\xi_i < \xi_{i+1} < \cdots < \xi_j, \quad j - i > i$$

must lie in the interval $R(\overline{s}')$. Now, since there are at least two roots ξ_i and $\xi_{i+1} \in R(\overline{s}')$, then, by Rolle's theorem, for some $\xi' \in R(\overline{s}')$ ($\xi_i < \xi' < \xi_{i+1}$), $f'(\xi') = 0$. Thus,

$$\left(\forall x \in R(\overline{s}') \right) \left[f'(x) = 0 \right] \Rightarrow f'(x) \equiv 0 \Rightarrow \deg(f) = 0,$$

a contradiction. $\square$

Corollary 8.5.10 *Consider two real roots ξ and ζ of a real univariate polynomial $f(x) \in \mathbb{R}[\curvearrowright]$ of positive degree $n > 0$. Then $\xi = \zeta$, if, for some $0 \leq m < n$, the following conditions hold:*

$$\begin{aligned} f^{(m)}(\xi) &= f^{(m)}(\zeta) = 0, \\ \operatorname{sgn}(f^{(m+1)}(\xi)) &= \operatorname{sgn}(f^{(m+1)}(\zeta)), \\ &\vdots \\ \operatorname{sgn}(f^{(n)}(\xi)) &= \operatorname{sgn}(f^{(n)}(\zeta)). \end{aligned}$$

PROOF.

Let

$$\overline{s}'' = \langle 0, \operatorname{sgn}(f^{(m+1)}(\xi)), \dots, \operatorname{sgn}(f^{(n)}(\xi)) \rangle,$$

and $R(\overline{s}'')$ be the sign-invariant region determined by $\overline{s}''$ with respect to $\overline{\text{FOURIER}}(f^{(m)})$. Since ξ and $\zeta \in R(\overline{s}'')$, we see that $R(\overline{s}'')$ is a nonempty interval, over which $f^{(m)}$ vanishes. Hence $f^{(m)}$ is identically zero. But this would contradict the fact that $\deg(f) = n > m \geq 0$. $\square$

Let us define $\operatorname{sgn}_\xi(\overline{\text{FOURIER}}(f))$ to be the sign sequence obtained by evaluating the polynomials of $\overline{\text{FOURIER}}(f)$ at ξ :

$$\begin{aligned} &\operatorname{sgn}_\xi(\overline{\text{FOURIER}}(f)) \\ &= \left\langle \operatorname{sgn}(f(\xi)), \operatorname{sgn}(f'(\xi)), \operatorname{sgn}(f^{(2)}(\xi)), \dots, \operatorname{sgn}(f^{(n)}(\xi)) \right\rangle. \end{aligned}$$

As an immediate corollary of Thom's lemma, we have:

Corollary 8.5.11 *Let ξ and ζ be two real roots of a real univariate polynomial $f(x) \in \mathbb{R}[\curvearrowright]$ of positive degree $n > 0$. Then $\xi = \zeta$, if the following condition holds:*

$$\operatorname{sgn}_\xi(\overline{\text{FOURIER}}(f')) = \operatorname{sgn}_\zeta(\overline{\text{FOURIER}}(f')). \quad \square$$

Representation of Real Algebraic Numbers

A real algebraic number $\alpha \in \mathbb{A}$ can be represented by its polynomial $f(x) \in \mathbb{Z}[\wedge]$, an integral polynomial with α as a root, and additional information identifying this particular root. Let

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0, \quad \deg(f) = n,$$

and assume that the distinct real roots of $f(x)$ have been enumerated as follows:

$$\alpha_1 < \alpha_2 < \cdots < \alpha_{j-1} < \alpha_j = \alpha < \alpha_{j+1} < \cdots < \alpha_l,$$

where $l \leq n = \deg(f)$.

While in certain cases (i.e., for input and output), we may require $f(x)$ to be a minimal polynomial α , in general, we relax this condition.

The Following is a list of possible representations of α :

1. **Order Representation:** The algebraic number α is represented as a pair consisting of its polynomial, f , and its index, j , in the sequence enumerating the real roots of f :

$$\langle \alpha \rangle_o = \langle f, j \rangle.$$

Clearly, this representation requires only $O(n \lg \|f\|_1 + \log n)$ bits.

2. **Sign Representation:** The algebraic number α is represented as a pair consisting of its polynomial, f , and a sign sequence, $\bar{s}$, representing the signs of the Fourier sequence of f' evaluated at the root α :

$$\langle \alpha \rangle_s = \langle f, \bar{s} = \text{sgn}_\alpha(\overline{\text{FOURIER}(f')}) \rangle.$$

The validity of this representation follows easily from the Little Thom's theorem. The sign representation requires only $O(n \lg \|f\|_1 + n)$ bits.

3. **Interval Representation:** The algebraic number α is represented as a triple consisting of its polynomial, f , and the two end points of an isolating interval, (l, r) ($l, r \in \mathbb{Q}$, $l < r$) containing only α :

$$\langle \alpha \rangle_i = \langle f, l, r \rangle.$$

By definition,

$$\max(\alpha_{j-1}, -1 - \|f\|_\infty) < l < \alpha_j = \alpha < r < \min(\alpha_{j+1}, 1 + \|f\|_\infty),$$

using our bounds for the real root separation, we see that the interval representation requires only $O(n \lg \|f\|_1 + n \lg n)$ bits.

Additionally, we require that the representation be *normalized* in the sense that if $\alpha \neq 0$ then $0 \notin (l, r)$, i.e., l and r have the same sign. We will provide a simple algorithm to normalize any arbitrary interval representation.

Example 8.5.8 Consider the representations of the following two algebraic numbers $-\sqrt{2} + \sqrt{3}$ and $\sqrt{2} + \sqrt{3}$:

$$\begin{aligned}
\langle -\sqrt{2} + \sqrt{3} \rangle_o &= \langle x^4 - 10x^2 + 1, 3 \rangle, \\
\langle \sqrt{2} + \sqrt{3} \rangle_o &= \langle x^4 - 10x^2 + 1, 4 \rangle, \\
\langle -\sqrt{2} + \sqrt{3} \rangle_s &= \langle x^4 - 10x^2 + 1, (-1, -1, +1) \rangle, \\
\langle \sqrt{2} + \sqrt{3} \rangle_s &= \langle x^4 - 10x^2 + 1, (+1, +1, +1) \rangle, \\
\langle -\sqrt{2} + \sqrt{3} \rangle_i &= \langle x^4 - 10x^2 + 1, 1/11, 1/2 \rangle, \\
\langle \sqrt{2} + \sqrt{3} \rangle_i &= \langle x^4 - 10x^2 + 1, 3, 7/2 \rangle. \quad \square
\end{aligned}$$

Here, we shall concentrate only on the interval representation, as it appears to be the best representation for a wide class of models of computation. While some recent research indicates that the sign representation leads to certain efficient parallel algebraic algorithms, it has yet to find widespread usage. Moreover, many of the key algebraic ideas are easier to explain for interval representation than the others. Henceforth, unless explicitly stated, we shall assume that the real algebraic numbers are given in the interval representation and are written without a subscript:

$$\langle \alpha \rangle = \langle f, l, r \rangle.$$

The following interval arithmetic operations simplify our later exposition:

Let $I_1 = (l_1, r_1) = \{x : l_1 < x < r_1\}$ and $I_2 = (l_2, r_2) = \{x : l_2 < x < r_2\}$ be two real intervals; then

$$\begin{aligned}
I_1 + I_2 &= (l_1 + l_2, r_1 + r_2) \\
&= \{x + y : l_1 < x < r_1 \text{ and } l_2 < y < r_2\},
\end{aligned}$$

$$\begin{aligned}
I_1 - I_2 &= (l_1 - r_2, r_1 - l_2) \\
&= \{x - y : l_1 < x < r_1 \text{ and } l_2 < y < r_2\},
\end{aligned}$$

$$\begin{aligned}
I_1 \cdot I_2 &= (\min(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2), \max(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2)) \\
&= \{xy : l_1 < x < r_1 \text{ and } l_2 < y < r_2\}.
\end{aligned}$$

We begin by describing algorithms for *normalization*, *refinement* and *sign evaluation*.

Normalization

NORMALIZE(α)

Input: A real algebraic number $\alpha = \langle f, l, r \rangle \in \mathbb{A}$.

Output: A representation of $\alpha = \langle f, l', r' \rangle$ such that $0 \notin (l', r')$.

```

 $p := 1/(1 + \|f\|_\infty);$ 

if  $\text{Var}_l(\overline{S}) > \text{Var}_{-p}(\overline{S})$  then
    return  $\alpha = \langle f, l, -p \rangle$ 
elseif  $\text{Var}_{-p}(\overline{S}) > \text{Var}_p(\overline{S})$  then
    return  $\alpha = 0$ 
else
    return  $\alpha = \langle f, p, r \rangle$ 
end{if };
end{NORMALIZE}    □

```

The correctness of the theorem is a straightforward consequence of Sturm's theorem and bounds on the nonzero zeros of f . It is easily seen that the algorithm requires $O(n^2)$ arithmetic operations.

Refinement

REFINE(α)

Input: A real algebraic number $\alpha = \langle f, l, r \rangle \in \mathbb{A}$.

Output: A finer representation of $\alpha = \langle f, l', r' \rangle$ such that $2(r' - l') \leq (r - l)$.

```

Let  $\overline{S} = \overline{\text{STURM}}(f, f')$  be a Sturm sequence;

 $m := (l + r)/2;$ 
if  $\text{Var}_l(\overline{S}) > \text{Var}_m(\overline{S})$  then
    return  $\langle f, l, m \rangle$ 
else
    return  $\langle f, m, r \rangle$ 
end{if };
end{REFINE}    □

```

Again the correctness of the algorithm follows from the Sturm's theorem and its time complexity is $O(n^2)$.

Sign Evaluation

Input: $\text{SIGN}(\alpha, g)$ A real algebraic number $\alpha = \langle f, l, r \rangle \in \mathbb{A}$, and a univariate rational polynomial $g(x) \in \mathbb{Q}[\curvearrowright]$.

Output: $\text{sgn}(g(\alpha)) = \text{sign of } g \text{ at } \alpha$.

Let $\overline{S}_g = \overline{\text{STURM}}(f, f'g)$, be a Sturm sequence;

return $\text{Var}_l(\overline{S}_g) - \text{Var}_r(\overline{S}_g)$;
end{SIGN} $\square$

The correctness of the algorithm follows from Sturm-Tarski theorem, since

$$\begin{aligned} & \text{Var}_l(\overline{\text{STURM}}(f, f'g)) - \text{Var}_r(\overline{\text{STURM}}(f, f'g)) \\ &= c_f \left[g > 0 \right]_l^r - c_f \left[g < 0 \right]_l^r \\ &= \begin{cases} +1, & \text{if } g(\alpha) > 0; \\ 0, & \text{if } g(\alpha) = 0; \\ -1, & \text{if } g(\alpha) < 0; \end{cases} \end{aligned}$$

and since f has only one root α in the interval (l, r) . The algorithm has a time complexity of $O(n^2)$. This algorithm has several applications: for instance, one can compare an algebraic number α with a rational number p/q by evaluating the sign of the polynomial $qx - p$ at α ; one can compute the multiplicity of a root α of a polynomial f by computing the signs of the polynomials $f', f^{(2)}, f^{(3)}$, etc., at α .

Conversion Among Representations

Interval representation to order representation:

INTERVALTOORDER(α)
Input: A real algebraic number $\langle \alpha \rangle_i = \langle f, l, r \rangle \in \mathbb{A}$.

Output: Its order representation $\langle \alpha \rangle_o = \langle f, j \rangle$.

Let $\overline{S} = \overline{\text{STURM}}(f, f')$ be a Sturm sequence;

return $\langle f, \text{Var}_{(-1-\|f\|_\infty)}(\overline{S}) - \text{Var}_r(\overline{S}) \rangle$;
end{INTERVALTOORDER} $\square$

Interval representation to sign representation:

```

INTERVALTOSIGN( $\alpha$ )
Input:    A real algebraic number  $\langle \alpha \rangle_i = \langle f, l, r \rangle \in \mathbb{A}$ .

Output:  Its sign representation  $\langle \alpha \rangle_s = \langle f, \bar{s} \rangle$ .

    Let  $\langle f', f^{(2)}, \dots, f^{(n)} \rangle = \overline{\text{FOURIER}}(f')$ ;

     $\bar{s} := \left\langle \begin{array}{l} \text{SIGN}(\alpha, f'), \\ \text{SIGN}(\alpha, f^{(2)}), \\ \vdots \\ \text{SIGN}(\alpha, f^{(n)}) \end{array} \right\rangle$ ;
    return  $\langle f, \bar{s} \rangle$ ;
end{INTERVALTOORDER}    □

```

Again, the correctness of these algorithms follow from Sturm-Tarski theorem. The algorithm INTERVALTOORDER has a time complexity of $O(n^2)$, and INTERVALTOSIGN has a complexity of $O(n^3)$.

Arithmetic Operations

Additive inverse:

```

ADDITIVEINVERSE( $\alpha$ )
Input:    A real algebraic number  $\langle \alpha \rangle = \langle f, l, r \rangle \in \mathbb{A}$ .

Output:   $-\alpha$  in its interval representation.

    return  $\langle f(-x), -r, -l \rangle$ ;
end{ADDITIVEINVERSE}    □

```

The correctness follows from the fact that if α is a root of $f(x)$, then $-\alpha$ is a root of $f(-x)$. Clearly, the algorithm has a linear time complexity.

Multiplicative inverse:

```

MULTIPLICATIVEINVERSE( $\alpha$ )
Input:    A nonzero real algebraic number  $\langle \alpha \rangle = \langle f, l, r \rangle \in \mathbb{A}$ .

Output:   $1/\alpha$  in its interval representation.

    return  $\left\langle x^{\deg(f)} f\left(\frac{1}{x}\right), \frac{1}{r}, \frac{1}{l} \right\rangle$ ;
end{MULTIPLICATIVEINVERSE}    □

```

The correctness follows from the fact that if α is a root of $f(x)$, then $1/\alpha$ is a root of $x^{\deg(f)} f(1/x)$. Again, the algorithm has a linear time complexity.

Addition:

```

ADDITION( $\alpha_1, \alpha_2$ )
Input:    Two real algebraic numbers
             $\langle \alpha_1 \rangle = \langle f_1, l_1, r_1 \rangle$  and  $\langle \alpha_2 \rangle = \langle f_2, l_2, r_2 \rangle \in \mathbb{A}$ .

Output:   $\alpha_1 + \alpha_2 = \alpha_3 = \langle f_3, l_3, r_3 \rangle$  in its interval representation.

 $f_3 := \text{Resultant}_y(f_1(x - y), f_2(y));$ 
 $\overline{S} := \overline{\text{STURM}}(f_3, f'_3);$ 

 $l_3 := l_1 + l_2;$ 
 $r_3 := r_1 + r_2;$ 

while  $\text{Var}_{l_3}(\overline{S}) - \text{Var}_{r_3}(\overline{S}) > 1$  loop

     $\langle f_1, l_1, r_1 \rangle := \text{REFINE}(\langle f_1, l_1, r_1 \rangle);$ 
     $\langle f_2, l_2, r_2 \rangle := \text{REFINE}(\langle f_2, l_2, r_2 \rangle);$ 

     $l_3 := l_1 + l_2;$ 
     $r_3 := r_1 + r_2;$ 
end{loop }

return  $\langle f_3, l_3, r_3 \rangle;$ 
end{ADDITION}     $\square$ 

```

The correctness of the algorithm follows from the main properties of the resultant and the repeated refinement process yielding an isolating interval. The resulting polynomial f_3 in this algorithm has the following size complexities:

$$\deg(f_3) \leq n_1 n_2,$$

$$\|f_3\|_1 \leq 2^{O(n_1 n_2)} \|f_1\|_1^{n_2} \|f_2\|_1^{n_1},$$

where $\deg(f_1) = n_1$ and $\deg(f_2) = n_2$. It can now be shown that the complexity of the algorithm is $O(n_1^3 n_2^4 \lg \|f_1\|_1 + n_1^4 n_2^3 \lg \|f_2\|_1)$.

Multiplication:

MULTIPLICATION(α_1, α_2)

Input: Two real algebraic numbers
 $\langle \alpha_1 \rangle = \langle f_1, l_1, r_1 \rangle$ and $\langle \alpha_2 \rangle = \langle f_2, l_2, r_2 \rangle \in \mathbb{A}$.

Output: $\alpha_1 \cdot \alpha_2 = \alpha_3 = \langle f_3, l_3, r_3 \rangle$ in its interval representation.

$$f_3 := \text{Resultant}_y \left(y^{\deg(f_1)} f_1 \left(\frac{x}{y} \right), f_2(y) \right);$$

$$\overline{S} := \overline{\text{STURM}}(f_3, f_3');$$

$$l_3 := \min(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2);$$

$$r_3 := \max(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2);$$

while $\text{Var}_{l_3}(\overline{S}) - \text{Var}_{r_3}(\overline{S}) > 1$ loop

$$\langle f_1, l_1, r_1 \rangle := \text{REFINE}(\langle f_1, l_1, r_1 \rangle);$$

$$\langle f_2, l_2, r_2 \rangle := \text{REFINE}(\langle f_2, l_2, r_2 \rangle);$$

$$l_3 := \min(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2);$$

$$r_3 := \max(l_1 l_2, l_1 r_2, r_1 l_2, r_1 r_2);$$

end{loop};

return $\langle f_3, l_3, r_3 \rangle$;
end{MULTIPLICATION} $\square$

The correctness of the multiplication algorithm can be proven as before.
The size complexities of the polynomial f_3 are:

$$\deg(f_3) \leq n_1 n_2,$$

$$\|f_3\|_1 \leq n_1 n_2 \|f_1\|_1^{n_2} \|f_2\|_1^{n_1},$$

where $\deg(f_1) = n_1$ and $\deg(f_2) = n_2$. Thus, the complexity of the algorithm is $O(n_1^3 n_2^4 \lg \|f_1\|_1 + n_1^4 n_2^3 \lg \|f_2\|_1)$.

8.6 Real Geometry

Definition 8.6.1 (Semialgebraic Sets) A subset $S \subseteq \mathbb{R}^\kappa$ is said to be a *semialgebraic set* if it can be determined by a set-theoretic expression of the following form:

$$S = \bigcup_{i=1}^m \bigcap_{j=1}^{l_i} \left\{ \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^\kappa : \text{sgn}(\mathcal{U}_{\sqcup, \mathfrak{J}}(\xi_{\mathfrak{K}}, \dots, \xi_{\mathfrak{K}})) = \sim \sqcup, \mathfrak{J} \right\},$$

where $f_{i,j}$'s are multivariate polynomials in $\mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_n]$:

$$f_{i,j}(x_1, \dots, x_n) \in \mathbb{R}[\wedge_{\mathbb{K}}, \dots, \wedge_{\mathbb{K}}], \quad \sqsupset = \mathbb{K}, \dots, \geq, \quad \sqsubset = \mathbb{K}, \dots, \leq \sqsupset,$$

and $s_{i,j}$'s are corresponding set of signs in $\{-1, 0, +1\}$. $\square$

Such sets arise naturally in solid modeling (as constructive solid geometric models), in robotics (as kinematic constraint relations on the possible configurations of a rigid mechanical system) and in computational algebraic geometry (as classical loci describing convolutes, evolutes and envelopes).

Note that the semialgebraic sets correspond to the minimal class of subsets of $\mathbb{R}^{\mathbb{K}}$ of the following forms:

1. Subsets defined by an algebraic *inequality*:

$$S = \left\{ \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^{\mathbb{K}} : \mathcal{U}(\xi_{\mathbb{K}}, \dots, \xi_{\mathbb{K}}) > \mathbb{K} \right\},$$

where $f(x_1, \dots, x_n) \in \mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_n]$.

2. Subsets closed under following set-theoretic operations, *complementation*, *union* and *intersection*. That is, if S_1 and S_2 are two semialgebraic sets, then so are

$$\begin{aligned} S_1^c &= \left\{ p = \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^{\mathbb{K}} : p \notin S_1 \right\}, \\ S_1 \cup S_2 &= \left\{ p = \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^{\mathbb{K}} : p \in S_1 \text{ or } p \in S_2 \right\}, \\ S_1 \cap S_2 &= \left\{ p = \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^{\mathbb{K}} : p \in S_1 \text{ and } p \in S_2 \right\}. \end{aligned}$$

After certain simplifications, we can also formulate a *semialgebraic set* in the following equivalent manner: it is a finite union of sets of the form shown below:

$$\begin{aligned} S &= \left\{ \langle \xi_1, \dots, \xi_n \rangle : \right. \\ &\quad g_1(\xi_1, \dots, \xi_n) = \dots = g_r(\xi_1, \dots, \xi_n) = 0, \\ &\quad \left. g_{r+1}(\xi_1, \dots, \xi_n) > 0, \dots, g_s(\xi_1, \dots, \xi_n) > 0 \right\}, \end{aligned}$$

where the $g_i(x_1, \dots, x_n) \in \mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_n]$ ($i = 1, \dots, s$).

It is also not hard to see that every propositional algebraic sentence composed of algebraic inequalities and Boolean connectives defines a semialgebraic set. In such sentences,

1. A constant is a real number;
2. An algebraic variable assumes a real number as its value; there are finitely many such algebraic variables: $x_1, x_2, \dots, x_n$.

3. An algebraic expression is a constant, or a variable, or an expression combining two algebraic expressions by an arithmetic operator: “+” (addition), “−” (subtraction), “.” (multiplication) and “/” (division).
4. An atomic Boolean predicate is an expression comparing two arithmetic expressions by a binary relational operator: “=” (equation), “≠” (inequation), “>” (strictly greater), “<” (strictly less), “≥” (greater than or equal to) and “≤” (less than or equal to).
5. A propositional sentence is an atomic Boolean predicate, a negation of a propositional sentence given by the unary Boolean connective: “¬” (negation), or a sentence combining two propositional sentences by a binary Boolean connective: “⇒” (implication), “∧” (conjunction) and “∨” (disjunction).

Thus another definition of semialgebraic sets can be given in terms of a propositional algebraic sentence $\psi(x_1, \dots, x_n)$ involving n algebraic variables: This is a subset $S \subseteq \mathbb{R}^\kappa$ determined by ψ as follows:

$$S = \left\{ \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^\kappa \quad : \quad \psi(\xi_\mu, \dots, \xi_\kappa) = \text{True} \right\}.$$

It turns out that if we enlarge our propositional sentences to include first-order universal and existential quantifiers (the so-called *Tarski sentences*), then the set determined by such sentences (*Tarski sets*) are also semialgebraic sets. That is, quantifiers do not add any additional power! This result follows from Tarski’s famous quantifier elimination theorem, which states that any set S determined by a quantified sentence is also determined by a quantifier-free sentence ψ' , or

Proposition 8.6.1 *Every Tarski set has a quantifier-free defining sentence.* $\square$

Later on in this section, we shall come back to a more detailed discussion of Tarski sentences and some results on an effective procedure to decide if a given Tarski sentence is true—geometric theorem proving problem.

Example 8.6.2 Some examples of such propositional algebraic sentences include the following:

1.

$$\phi_1(x_1, x_2, \dots, x_n) = x_1^2 + x_2^2 + \dots + x_n^2 < 0.$$

The semialgebraic set defined by ϕ_1 is the empty set.

2.

$$(x^4 - 10x^2 + 1 = 0) \wedge (2x < 7) \wedge (x > 3)$$

defines the only root of the polynomial $x^4 - 10x^2 + 1$ in the isolating interval $[3, 7/2]$ —namely, the real algebraic number $\sqrt{2} + \sqrt{3}$.

3.

$$(x^2 + bx + c = 0) \wedge (y^2 + by + c = 0) \wedge (x \neq y),$$

which has a real solution $\langle x, y \rangle$ if and only if $b^2 > 4c$, i.e., when the quadratic polynomial $x^2 + bx + c$ has two distinct real roots. $\square$

The following properties of semialgebraic sets are noteworthy:

- A semialgebraic set S is *semialgebraically connected*, if it is not the union of two disjoint nonempty semialgebraic sets.

A semialgebraic set S is *semialgebraically path connected* if for every pair of points $p, q \in S$ there is a semialgebraic path connecting p and q (one-dimensional semialgebraic set containing p and q) that lies in S .

- A semialgebraic set is semialgebraically connected if and only if it is semialgebraically path connected. Working over the real numbers, it can be seen that a semialgebraic set is semialgebraically path connected if and only if it is *path connected*. Thus, we may say a semialgebraic set is connected when we mean any of the preceding notions of connectedness.
- A *connected component* (semialgebraically connected component) of a semialgebraic set S is a maximal (semialgebraically) connected subset of S .
- Every semialgebraic set has a finite number of connected components.
- If S is a semialgebraic set, then its *interior*, $\text{int}(S)$, *closure*, $\overline{S}$, and *boundary* $\partial(S) = \overline{S} \setminus \text{int}(S)$ are all semialgebraic.
- For any semialgebraic subset $S \subseteq \mathbb{R}^n$, a *semialgebraic decomposition* of S is a finite collection $\mathcal{K}$ of disjoint connected semialgebraic subsets of S whose union is S . Every semialgebraic set admits a semialgebraic decomposition.

Let $\mathcal{F} = \{f_{i,j} : i = 1, \dots, m, j = 1, \dots, l_i\} \subseteq \mathbb{R}[x_1, \dots, x_n]$ be a set of real multivariate polynomials in n variables. Any point $p = \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^n$ has a *sign assignment* with respect to $\mathcal{F}$ as follows:

$$\text{sgn}_{\mathcal{F}}(p) = \left\langle \text{sgn}(f_{i,j}(\xi_1, \dots, \xi_n)) : i = 1, \dots, m, j = 1, \dots, l_i \right\rangle.$$

Using sign assignments, we can define the following equivalence relation: Given two points $p, q \in \mathbb{R}^n$, we say

$$p \sim_{\mathcal{F}} q, \quad \text{if and only if} \quad \text{sgn}_{\mathcal{F}}(p) = \text{sgn}_{\mathcal{F}}(q).$$

Now consider the partition of $\mathbb{R}^\kappa$ defined by the equivalence relation $\sim_{\mathcal{F}}$; each equivalence class is a semialgebraic set comprising finitely many connected semialgebraic components. Each such equivalence class is called a *sign class* of $\mathcal{F}$.

Clearly, the collection of semialgebraic components of all the sign classes, $\mathcal{K}$, provides a semialgebraic decomposition of $\mathbb{R}^\kappa$. Furthermore, if $S \subseteq \mathbb{R}^\kappa$ is a semialgebraic set defined by some subset of $\mathcal{F}$, then it is easily seen that

$$\{C \in \mathcal{K} : C \cap S \neq \emptyset\},$$

defines a semialgebraic decomposition of the semialgebraic set S .

- A *semialgebraic cell-complex* (*cellular decomposition*) for $\mathcal{F}$ is a semialgebraic decomposition of $\mathbb{R}^\kappa$ into finitely many disjoint semialgebraic subsets, $\{C_i\}$, called *cells* such that we have the following:

1. Each cell C_i is homeomorphic to $\mathbb{R}^{\delta(i)}$, $0 \leq \delta(i) \leq n$. $\delta(i)$ is called the *dimension of the cell* C_i , and C_i is called a $\delta(i)$ -*cell*.
2. Closure of each cell C_i , $\overline{C_i}$, is a union of some cells C_j 's:

$$\overline{C_i} = \bigcup_j C_j.$$

3. Each C_i is contained in some semialgebraic sign class of $\mathcal{F}$ —that is, the sign of each $f_{i,j} \in \mathcal{F}$ is *invariant* in each C_i .

Subsequently, we shall study a particularly “nice” semialgebraic cell-complex that is obtained by Collin’s *cylindrical algebraic decomposition* or CAD.

8.6.1 Real Algebraic Sets

A special class of semialgebraic sets are the *real algebraic sets* determined by a conjunction of algebraic equalities.

Definition 8.6.3 (Real Algebraic Sets) A subset $Z \subseteq \mathbb{R}^\kappa$ is said to be a *real algebraic set*, if it can be determined by a system of algebraic equations as follows:

$$Z = \left\{ \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^\kappa : \mathcal{U}_{\mathbb{K}}(\xi_{\mathbb{K}}, \dots, \xi_{\mathbb{K}}) = \dots = \mathcal{U}_{\mathbb{J}}(\xi_{\mathbb{K}}, \dots, \xi_{\mathbb{K}}) = \mathbb{K} \right\},$$

where f_i 's are multivariate polynomials in $\mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_n]$. $\square$

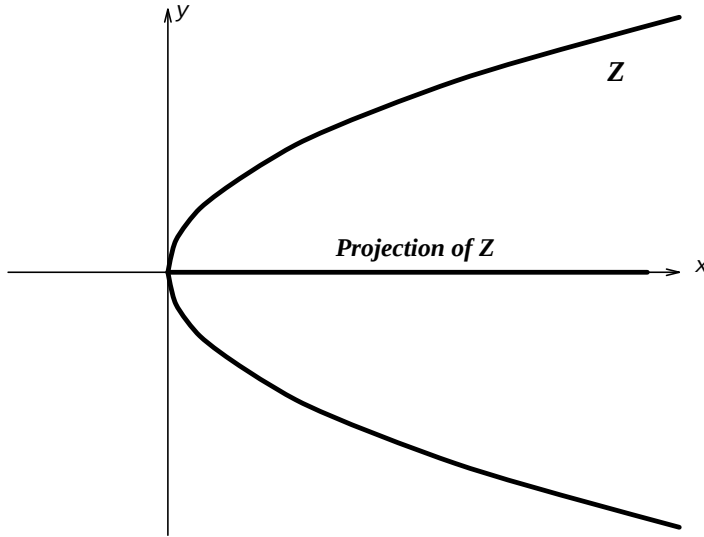


Figure 8.1: Projection of a real algebraic set Z .

Note that a real algebraic set Z could have been defined by a single algebraic equation as follows:

$$Z = \left\{ \langle \xi_1, \dots, \xi_n \rangle \in \mathbb{R}^n : \mathcal{U}_{\neq}(\xi_{\neq}, \dots, \xi_{\neq}) + \dots + \mathcal{U}_{>}(\xi_{\neq}, \dots, \xi_{\neq}) = \neq \right\},$$

as we are working over the field of reals, $\mathbb{R}$.

While real algebraic sets are quite interesting for the same reasons as complex algebraic varieties, they lack certain “nice” geometric properties and hence, are somewhat unwieldy. For instance, *real algebraic sets are not closed under projection onto a subspace*.

Consider the following simple real algebraic set defining a parabola:

$$Z = \left\{ \langle x, y \rangle \in \mathbb{R}^2 : \curvearrowright = \curvearrowright^{\neq} \right\}.$$

If π_x is a projection map defined as follows:

$$\begin{aligned} \pi &: \mathbb{R}^2 \rightarrow \mathbb{R} \\ &: \langle x, y \rangle \mapsto x, \end{aligned}$$

then

$$\pi(Z) = \{x \in \mathbb{R} : \curvearrowright \geq \neq\}.$$

See Figure 8.1.

Clearly, $\pi(Z)$ is not algebraic, since only algebraic sets in $\mathbb{R}$ are finite or entire $\mathbb{R}$. However, it is semialgebraic. Additionally, we shall see that

semialgebraic sets are closed under projection as well as various other set-theoretic operations. In fact, semialgebraic sets are the smallest class of subsets of $\mathbb{R}^\kappa$ containing real algebraic sets and closed under projection. In the next two subsection, we shall develop some machinery that among other things shows that semialgebraic sets are closed under projection.

8.6.2 Delineability

Let $f_i(x_1, \dots, x_{n-1}, x_n) \in \mathbb{R}[\curvearrowright, \dots, x_{n-1}, x_n]$ be a polynomial in n variables:

$$\begin{aligned} f_i(x_1, \dots, x_{n-1}, x_n) &= f_i^{d_i}(x_1, \dots, x_{n-1}) x_n^{d_i} \\ &\quad + \dots + f_i^0(x_1, \dots, x_{n-1}), \end{aligned}$$

where f_i^j 's are in $\mathbb{R}[\curvearrowright, \dots, x_{n-1}]$. Let $p' = \langle \xi_1, \dots, \xi_{n-1} \rangle \in \mathbb{R}^{\kappa - \mu}$. Then we write

$$f_{i,p'}(x_n) = f_i^{d_i}(p') x_n^{d_i} + \dots + f_i^0(p'),$$

for the univariate polynomial obtained by substituting p' for the first $(n-1)$ variables.

Definition 8.6.4 (Delineable Sets) Let

$$\mathcal{F} = \left\{ \begin{array}{l} f_1(x_1, \dots, x_n), \\ f_2(x_1, \dots, x_n), \\ \vdots \\ f_s(x_1, \dots, x_n) \end{array} \right\} \subseteq \mathbb{R}[x_1, \dots, x_n]$$

be a set of s n -variate real polynomials. Let $C \subseteq \mathbb{R}^{\kappa - \mu}$ be a nonempty set homeomorphic to $\mathbb{R}^\delta$ ($0 \leq \delta \leq n-1$).

We say $\mathcal{F}$ is *delineable* on C (or, C is $\mathcal{F}$ -*delineable*), if it satisfies the following invariant properties:

1. For every $1 \leq i \leq s$, the *total number of complex roots* of $f_{i,p'}$ (counting multiplicity) remains invariant as p' varies over C .
2. For every $1 \leq i \leq s$, the *number of distinct complex roots* of $f_{i,p'}$ (not counting multiplicity) remains invariant as p' varies over C .
3. For every $1 \leq i < j \leq s$, the *total number of common complex roots* of $f_{i,p'}$ and $f_{j,p'}$ (counting multiplicity) remains invariant as p' varies over C . $\square$

Theorem 8.6.2 *Let $\mathcal{F} \subseteq \mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_n]$ be a set of polynomials as in the preceding definition, and let $C \subseteq \mathbb{R}^{\mathbb{K}-\mathbb{K}}$ be a connected maximal $\mathcal{F}$ -delineable set. Then C is semialgebraic.*

PROOF.

We show that all the three invariant properties of the definition for delinability have semialgebraic characterizations.

(1) The first condition states that

$$\left(\forall i \right) \left(\forall p' \in C \right) \left[|\mathcal{Z}(f_{i,p'})| = \text{invariant} \right],$$

where $\mathcal{Z}(f)$ denotes the complex roots of f . This condition is simply equivalent to saying that “ $\deg(f_{i,p'})$ is invariant (say, k_i).” A straightforward semialgebraic characterization is as follows:

$$\left(\forall 1 \leq i \leq s \right) \left(\exists 0 \leq k_i \leq d_i \right) \\ \left[(\forall k > k_i) [f_i^k(x_1, \dots, x_{n-1}) = 0] \wedge f_i^{k_i}(x_1, \dots, x_{n-1}) \neq 0 \right]$$

holds for all $p' \in C$.

(2) The second condition, in view of the first condition, can be restated as follows:

$$\left(\forall i \right) \left(\forall p' \in C \right) \left[|\mathcal{CZ}(f_{i,p'}, D_{x_n}(f_{i,p'}))| = \text{invariant} \right],$$

where D_{x_n} denotes the formal derivative operator with respect to the variable x_n and $\mathcal{CZ}(f, g)$ denotes the common complex roots of f and g . Using principal subresultant coefficients, we can provide the following semialgebraic characterization:

$$\left(\forall 1 \leq i \leq s \right) \left(\exists 0 \leq l_i \leq d_i - 1 \right) \\ \left[(\forall l < l_i) [\text{PSC}_l^{x_n}(f_i(x_1, \dots, x_n), D_{x_n}(f_i(x_1, \dots, x_n))) = 0] \right. \\ \left. \wedge \text{PSC}_{l_i}^{x_n}(f_i(x_1, \dots, x_n), D_{x_n}(f_i(x_1, \dots, x_n))) \neq 0 \right]$$

holds for all $p' \in C$; here $\text{PSC}_l^{x_n}$ denotes the l^{th} principal subresultant coefficient with respect to x_n .

(3) Finally, the last condition can be restated as follows:

$$\left(\forall i \neq j \right) \left(\forall p' \in C \right) \left[|\mathcal{CZ}(f_{i,p'}, f_{j,p'})| = \text{invariant} \right].$$

Using principal subresultant coefficients, we can provide the following semialgebraic characterization:

$$\left(\forall 1 \leq i < j \leq s \right) \left(\exists 0 \leq m_{ij} \leq \min(d_i, d_j) \right) \\ \left[(\forall m < m_{ij}) [\text{PSC}_m^{x_n}(f_i(x_1, \dots, x_n), f_j(x_1, \dots, x_n)) = 0] \right. \\ \left. \wedge \text{PSC}_{m_{ij}}^{x_n}(f_i(x_1, \dots, x_n), f_j(x_1, \dots, x_n)) \neq 0 \right],$$

holds for all $p' \in C$; here $\text{PSC}_m^{x_n}$ denotes the m^{th} principal subresultant coefficient with respect to x_n . $\square$

In summary, given a set of polynomials, $\mathcal{F} \in \mathbb{R}[\curvearrowright, \dots, x_n]$, as shown below, we can compute another set of $(n-1)$ -variate polynomials, $\Phi(\mathcal{F}) \in \mathbb{R}[\curvearrowright, \dots, x_{n-1}]$, which precisely characterizes the connected maximal $\mathcal{F}$ -delineable subsets of $\mathbb{R}^{\times-\mathbb{K}}$. Let

$$\mathcal{F} = \{f_1, f_2, \dots, f_s\};$$

then

$$\begin{aligned} \Phi(\mathcal{F}) = & \left\{ f_i^k(x_1, \dots, x_{n-1}) : 1 \leq i \leq s, 0 \leq k \leq d_i \right\} \\ & \cup \left\{ \text{PSC}_l^{x_n}(f_i(x_1, \dots, x_n), D_{x_n}(f_i(x_1, \dots, x_n))) : \right. \\ & \quad \left. 1 \leq i \leq s, 0 \leq l \leq d_i - 1 \right\} \\ & \cup \left\{ \text{PSC}_m^{x_n}(f_i(x_1, \dots, x_n), f_j(x_1, \dots, x_n)) : \right. \\ & \quad \left. 1 \leq i < j \leq s, 0 \leq m \leq \min(d_i, d_j) \right\}. \end{aligned}$$

Now, we come to the next important property that *delineability* provides. Clearly, by definition, *the total number of distinct complex roots of the set of polynomials $\mathcal{F}$ is invariant over the connected set $C \subseteq \mathbb{R}^{\times-\mathbb{K}}$* . But it is also true that *the total number of distinct real roots of $\mathcal{F}$ is invariant over the set C* .

Consider an arbitrary polynomial $f_i \in \mathcal{F}$; since it has real coefficients, its complex roots must occur in conjugate pairs. Thus as $f_{i,p'}$ varies to $f_{i,q'}$ such that some pair of complex conjugate roots (which are necessarily distinct) coalesce into a real root (of multiplicity two), somewhere along a path from p' to q' the total number of distinct roots of f must have dropped. Thus a transition from a nonreal root to a real root is impossible over C . Similar arguments also show that a transition from a real root to a nonreal is impossible as it would imply a splitting of a real root into a pair of distinct complex conjugate roots.

More formally, we argue as follows:

Lemma 8.6.3 *Let $\mathcal{F} \subseteq \mathbb{R}[\curvearrowright, \dots, x_n]$ be a set of polynomials as before, and let $C \subseteq \mathbb{R}^{\times-\mathbb{K}}$ be a connected $\mathcal{F}$ -delineable set. Then the total number of distinct real roots of $\mathcal{F}$ is locally invariant over the set C .*

PROOF.

Consider a polynomial $f_i \in \mathcal{F}$. Let p' and q' be two points in C such that $\|p' - q'\| < \epsilon$ and assume that every root of $f_{i,p'}$ differs from some root of $f_{i,q'}$ by no more than

$$\delta < \left(\frac{1}{2}\right) \text{Separation}(f_{i,p'}),$$

where Separation denotes the complex root separation of $f_{i,p'}$.

Now, let z_j be a complex root of $f_{i,p'}$ such that a disc of radius δ centered around z_j in the complex plane contains a real root y_j of $f_{i,q'}$. But then it can also be shown that y_j is also in a disc of radius δ about $\bar{z}_j$, the complex conjugate of z_j . But then z_j and $\bar{z}_j$ would be closer to each other than Separation($f_{i,p'}$), contradicting our choice of δ . Thus the total number of distinct complex roots as well as the total number of distinct real roots of f_i remains invariant in any small neighborhood of a point $p' \in C$. $\square$

Lemma 8.6.4 *Let $\mathcal{F} \subseteq \mathbb{R}[\curvearrowright, \dots, x_n]$ and $C \subseteq \mathbb{R}^{\kappa-\mu}$ be a connected $\mathcal{F}$ -delineable set, as in the previous lemma. Then the total number of distinct real roots of $\mathcal{F}$ is invariant over the set C .*

PROOF.

Let p' and q' be two arbitrary points in C connected by a path $\gamma : [0, 1] \rightarrow C$ such that $\gamma(0) = p'$ and $\gamma(1) = q'$. Since γ can be chosen to be continuous (even semialgebraic) the image of the compact set $[0, 1]$ under γ , $\Gamma = \gamma([0, 1])$ is also compact. At every point $r' \in \Gamma$ there is a small neighborhood $N(r')$ over which the total number of distinct real roots of $\mathcal{F}$ remains invariant. Now, since the path Γ has a finite cover of such neighborhoods $N(r'_1), N(r'_2), \dots, N(r'_k)$ over each of which the total number of distinct real roots remain invariant, this number also remains invariant over the entire path Γ . Hence, as C is path connected, the lemma follows immediately. $\square$

As an immediate corollary of the preceding lemmas we have the following:

Corollary 8.6.5 *Let $\mathcal{F} \subseteq \mathbb{R}[\curvearrowright, \dots, x_n]$ be a set of polynomials, delineable on a connected set $C \subseteq \mathbb{R}^{\kappa-\mu}$.*

1. *The complex roots of $\mathcal{F}$ vary continuously over C .*
2. *The real roots of $\mathcal{F}$ vary continuously over C , while maintaining their order; i.e., the j^{th} smallest real root of $\mathcal{F}$ varies continuously over C .*

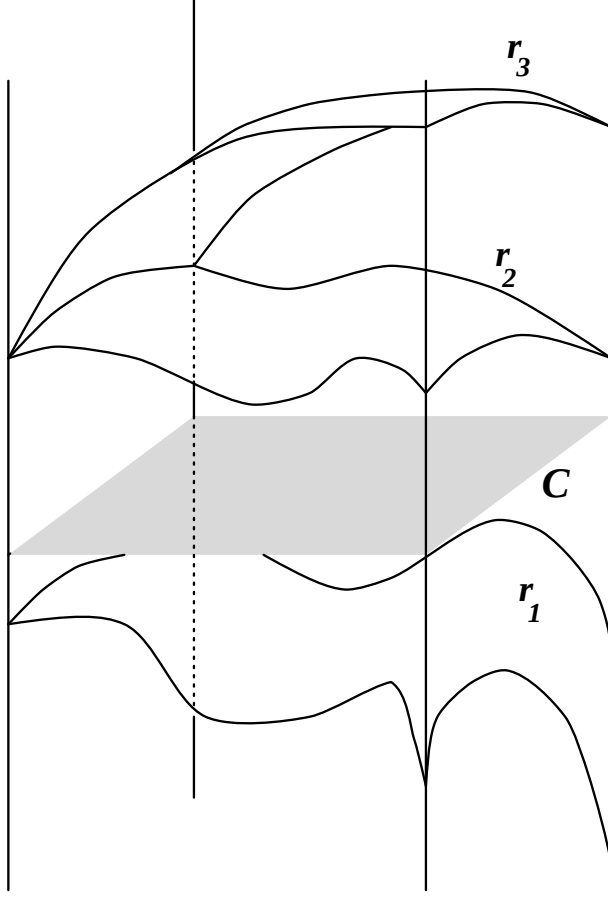
$\square$

Using this corollary, we can describe how the real roots are structured above C . Consider the cylinder over C obtained by taking the direct product of C with the two-point compactification of the reals, $\mathbb{R} \cup \{\pm\infty\}$. Note that the two-point compactification makes it possible to deal with vertical asymptotes of the real hypersurfaces defined by $\mathcal{F}$.

The cylinder $C \times (\mathbb{R} \cup \{\pm\infty\})$ can be partitioned as follows:

Definition 8.6.5 (Sections and Sectors) Suppose $\mathcal{F} \subseteq \mathbb{R}[\curvearrowright, \dots, x_n]$ is delineable on a connected set $C \subseteq \mathbb{R}^{\kappa-\mu}$. Assume that $\mathcal{F}$ has finitely many distinct real roots over C , given by m continuous functions

$$r_1(p'), r_2(p'), \dots, r_m(p'),$$

Figure 8.2: Sectors and sections over C .

where r_j denotes the j^{th} smallest root of $\mathcal{F}$. (See Figure 8.2). Then we have the following:

1. The j^{th} $\mathcal{F}$ -section over C is

$$\left\{ \langle p', x_n \rangle : p' \in C, x_n = r_j(p') \right\}.$$

2. The j^{th} ($0 < j < n$) intermediate $\mathcal{F}$ -sector over C is

$$\left\{ \langle p', x_n \rangle : p' \in C, r_j(p') < x_n < r_{j+1}(p') \right\}.$$

The lower semiinfinite $\mathcal{F}$ -sector over C is

$$\left\{ \langle p', x_n \rangle : p' \in C, x_n < r_1(p') \right\}.$$

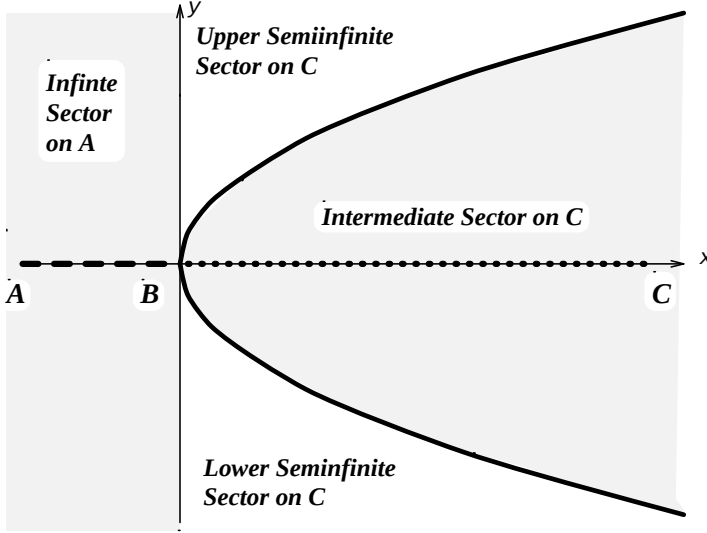


Figure 8.3: Sectors and sections for the parabola example.

The *upper semiinfinite* $\mathcal{F}$ -sector over C is

$$\left\{ \langle p', x_n \rangle : p' \in C, x_n > r_m(p') \right\}. \quad \square$$

Example 8.6.6 Consider the real polynomial $f(x, y) = y^2 - x \in \mathbb{R}[\curvearrowright, y]$ defining a parabola in the plane (see Figure 8.3). Note that

$$f(x, y) = y^2 (1) + y^1 (0) + y^0 (-x),$$

and thus

$$f^2(x) = 1, \quad f^1(x) = 0, \quad \text{and} \quad f^0(x) = -x.$$

Since the subresultant chain of $f = y^2 - x$ and $f' = D_y(f) = 2y$ are given by

$$\begin{aligned} \text{SubRes}_2(f, f') &= y^2 - x \\ \text{SubRes}_1(f, f') &= 2y \\ \text{SubRes}_0(f, f') &= -4x, \end{aligned}$$

the principal subresultant coefficients are

$$\text{PSC}_2(x) = 1, \quad \text{PSC}_1 = 2, \quad \text{and} \quad \text{PSC}_0 = -4x.$$

Thus

$$\Phi(\{f\}) = \{1, x\},$$

and the maximal connected f -delineable sets are $A = [-\infty, 0)$, $B = [0, 0]$ and $C = (0, +\infty]$. There is only one infinite sector over A , as for every $x \in A$, $y^2 - x$ has no real zero. There are two semiinfinite sectors and one section over B , as $y^2 - x$ has one zero (of multiplicity two) at $y = 0$. Finally, there are three sectors and two sections over C , as for every $x \in C$, $y^2 - x$ has two distinct real zeros (each of multiplicity one).

Note that as we traverse along the x -axis from $-\infty$ to $+\infty$, we see that $y^2 - x$ has two distinct complex zeros for all $x < 0$, which coalesce into one real zero at $x = 0$ and then split into two distinct real zeros for $x > 0$. $\square$

Observe that the $\mathcal{F}$ -sections and sectors are uniquely defined by the distinct real root functions of $\mathcal{F}$:

$$r_1(p'), r_2(p'), \dots, r_m(p'),$$

where it is implicitly assumed that not all $f_{i,p'} \equiv 0$ ($f_i \in \mathcal{F}$). It is sometimes easier to use a single multivariate polynomial $g = \Pi(\mathcal{F})(x_1, \dots, x_n)$ with

$$g'_p(x_n) = g(p', x_n)$$

vanishing precisely at the distinct roots $r_1(p'), r_2(p'), \dots, r_m(p')$.

$$\Pi(\mathcal{F})(x_1, \dots, x_n) = \prod_{f_i \in \mathcal{F}, f_{i,p'} \neq 0} f_i(x_1, \dots, x_n),$$

where $p' \in C$, a connected $\mathcal{F}$ -delineable set. By convention, we shall have $\Pi(\mathcal{F}) = 1$, if all $f_{i,p'} \equiv 0$ ($f_i \in \mathcal{F}$). Also, when $\Pi(\mathcal{F}) = \text{constant}$, the cylinder over C will have exactly one *infinite $\mathcal{F}$ -sector*: $C \times \{\mathbb{R} \cup \{\pm\infty\}\}$.

8.6.3 Tarski-Seidenberg Theorem

As an immediate consequence of the preceding discussions, we are now ready to show that *semialgebraic sets are closed under projection*. A more general result in this direction is the famous *Tarski-Seidenberg theorem*.

Definition 8.6.7 (Semialgebraic Map) A map $\psi : S \rightarrow T$, from a semialgebraic set $S \subseteq \mathbb{R}^>$ to a semialgebraic set $T \subseteq \mathbb{R}^\times$ is said to be a *semialgebraic map*, if its graph

$$\left\{ \langle s, \psi(s) \rangle \in \mathbb{R}^{>+\times} : s \in S \right\}$$

is a semialgebraic set in $\mathbb{R}^{>+\times}$. $\square$

Theorem 8.6.6 (Tarski-Seidenberg Theorem) Let S be a semialgebraic set in $\mathbb{R}^>$ and

$$\psi : \mathbb{R}^> \rightarrow \mathbb{R}^\times$$

be a semialgebraic map; then $\psi(S)$ is semialgebraic in $\mathbb{R}^\kappa$.

PROOF.

The proof is by induction on m . We start by considering the base case $m = 1$. Then the graph of ψ , say V , is a semialgebraic set in $\mathbb{R}^{\kappa+\mathbb{K}}$ and the image of S , $\psi(S)$, is a subset of $\mathbb{R}^\kappa$. After suitable renaming of the coordinates, we can so arrange that V is defined by a set of polynomials $\mathcal{F} \subseteq \mathbb{R}[\wedge_{\mathbb{K}}, \dots, x_{n+1}]$ and $\psi(S) = \pi(V)$ is the projection of V onto the first n coordinates:

$$\begin{aligned} \pi &: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^\kappa \\ &: \langle \xi_1, \dots, \xi_n, \xi_{n+1} \rangle \mapsto \langle \xi_1, \dots, \xi_n \rangle. \end{aligned}$$

Corresponding to the set $\mathcal{F}$, we can define the set of polynomials $\Phi(\mathcal{F})$ as earlier. Now, note that if C is a cell of a sign-invariant cell decomposition, $\mathcal{K}$ of $\mathbb{R}^\kappa$, defined by $\Phi(\mathcal{F})$, then C is a maximal connected $\mathcal{F}$ -delineable set.

Next, we claim that for every $C \in \mathcal{K}$,

$$C \cap \pi(V) \neq \emptyset \Rightarrow C \subseteq \pi(V).$$

To see this, note that since $C \cap \pi(V) \neq \emptyset$, there is a point $p \in V$, such that $p' = \pi(p) \in C$. Thus p belongs to some $\mathcal{F}$ -section or sector defined by some real functions $r_i(p')$ and $r_{i+1}(p')$. Now consider an arbitrary point $q' \in C$; since C is path connected, there is a path $\gamma' : [0, 1] \rightarrow C$ such that $\gamma'(0) = p'$ and $\gamma'(1) = q'$. This path can be lifted to a path in V , by defining $\gamma : [0, 1] \rightarrow V$ as follows:

$$\gamma(t) = \frac{r_{i+1}(\gamma'(t))[p - r_i(p')] - r_i(\gamma'(t))[p - r_{i+1}(p')]}{r_{i+1}(p') - r_i(p')},$$

where $t \in [0, 1]$.

Clearly, the path $\gamma([0, 1]) \in V$; $\pi(\gamma(t)) = \gamma'(t)$, and $q' \in \pi(V)$, as required. Hence $\psi(S) = \pi(V)$ can be expressed as a union of finitely many semialgebraic cells of the decomposition $\mathcal{K}$, since

$$\pi(V) \subseteq \bigcup \{C : C \cap \pi(V) \neq \emptyset\} \subseteq \pi(V).$$

Hence, $\psi(S)$ is semialgebraic in $\mathbb{R}^\kappa$.

For $m > 1$, the proof proceeds by induction, as any projection from $\Pi : \mathbb{R}^> \times \mathbb{R}^\kappa \rightarrow \mathbb{R}^\kappa$ can be expressed as a composition of the following two projection maps: $\Pi' : \mathbb{R}^{>-\mathbb{K}} \times \mathbb{R}^{\kappa+\mathbb{K}} \rightarrow \mathbb{R}^{\kappa+\mathbb{K}}$ and $\pi' : \mathbb{R}^{\kappa+\mathbb{K}} \rightarrow \mathbb{R}^\kappa$. $\square$

Corollary 8.6.7 *Let S be a semialgebraic set in $\mathbb{R}^>$ and*

$$\psi : \mathbb{R}^> \rightarrow \mathbb{R}^\kappa$$

be a polynomial map; then $\psi(S)$ is semialgebraic in $\mathbb{R}^\kappa$.

PROOF.

Let ψ be given by the following sequence of polynomials

$$g_k(x_1, \dots, x_m) \in \mathbb{R}[\curvearrowright, \curvearrowleft, \dots, \curvearrowright, \curvearrowleft], \quad \curvearrowright = \curvearrowright, \dots, \curvearrowleft.$$

Then the graph of the map is defined by

$$(S \times \mathbb{R}^\kappa) \cap \mathbb{T},$$

where

$$T = \left\{ \langle \xi_1, \dots, \xi_m, \zeta_1, \dots, \zeta_n \rangle \in \mathbb{R}^{>+\kappa} : \right. \\ \left. g_k(\xi_1, \dots, \xi_m) - \zeta_k = 0, \quad \text{for all } k = 1, \dots, n \right\}.$$

Thus ψ is a semialgebraic map and the rest follows from the Tarski-Seidenberg theorem. $\square$

8.6.4 Representation and Decomposition of Semialgebraic Sets

Using the ideas developed in this chapter (i.e., Sturm's theory and real algebraic numbers), we can already see how semialgebraic sets in $\mathbb{R} \cup \{\pm\infty\}$ can be represented and manipulated easily. In this one-dimensional case, the semialgebraic sets can be represented as a union of finitely many intervals whose endpoints are real algebraic numbers. For instance, given a set of univariate defining polynomials:

$$\mathcal{F} = \left\{ f_{i,j}(x) \in \mathbb{Q}[\curvearrowright] : \curvearrowright = \curvearrowright, \dots, \curvearrowright, \curvearrowleft = \curvearrowright, \dots, \curvearrowleft \right\},$$

we may enumerate all the real roots of the $f_{i,j}$'s (i.e., the real roots of the single polynomial $F = \prod_{i,j} f_{i,j}$) as

$$-\infty < \xi_1 < \xi_2 < \dots < \xi_{i-1} < \xi_i < \xi_{i+1} < \dots < \xi_s < +\infty,$$

and consider the following finite set $\mathcal{K}$ of elementary intervals defined by these roots:

$$[-\infty, \xi_1), [\xi_1, \xi_1], (\xi_1, \xi_2), \dots, \\ (\xi_{i-1}, \xi_i), [\xi_i, \xi_i], (\xi_i, \xi_{i+1}), \dots, [\xi_s, \xi_s], (\xi_s, +\infty].$$

Note that, these intervals are defined by real algebraic numbers with defining polynomial

$$\Pi(\mathcal{F}) = \prod_{f_{i,j} \neq 0 \in \mathcal{F}} f_{i,j}(x).$$

Now, any semialgebraic set in $S \subseteq \mathbb{R} \cup \{\pm\infty\}$ defined by $\mathcal{F}$:

$$S = \bigcup_{i=1}^m \bigcap_{j=1}^{l_i} \left\{ \xi \in \mathbb{R} \cup \{\pm\infty\} : \text{sgn}(\mathcal{U}_{\square, \mathbf{j}}(\xi)) = \sim_{\square, \mathbf{j}} \right\},$$

where $s_{i,j} \in \{-1, 0, +1\}$, can be seen to be the union of a subset of elementary intervals in $\mathcal{K}$. Furthermore, this subset can be identified as follows: if, for every interval $C \in \mathcal{K}$, we have a sample point $\alpha_C \in C$, then C belongs to S if and only if

$$\left(\forall i, j \right) \left[\text{sgn}(f_{i,j}(\alpha)) = s_{i,j} \right].$$

For each interval $C \in \mathcal{K}$, we can compute a sample point (an algebraic number) α_C as follows:

$$\alpha_C = \begin{cases} \xi_1 - 1, & \text{if } C = [-\infty, \xi_1); \\ \xi_i, & \text{if } C = [\xi_i, \xi_i]; \\ (\xi_i + \xi_{i+1})/2, & \text{if } C = (\xi_i, \xi_{i+1}); \\ \xi_s + 1, & \text{if } C = (\xi_s, +\infty]. \end{cases}$$

Note that the computation of the sample points in the intervals, their representations, and the evaluation of other polynomials at these points can all be performed by the Sturm theory developed earlier.

A generalization of the above representation to higher dimensions can be provided by using the machinery developed for delineability. In order to represent a semialgebraic set $S \subseteq \mathbb{R}^\kappa$, we may assume recursively that we can represent its projection $\pi(S) \subseteq \mathbb{R}^{\kappa-\mu}$ (also a semialgebraic set), and then represent S as a union of the sectors and sections in the cylinders above each cell of a semialgebraic decomposition of $\pi(S)$. This also leads to a semialgebraic decomposition of S .

We can further assign an algebraic sample point in each cell of the decomposition of S recursively as follows: Assume that the (algebraic) sample points for each cell of $\pi(S)$ have already been computed recursively. Note that a vertical line passing through a sample point of $\pi(S)$ intersects the sections above the corresponding cell at algebraic points. From these algebraic points, we can derive the algebraic sample points for the cells of S , in a manner similar to the one-dimensional case.

If $\mathcal{F}$ is a defining set for $S \subseteq \mathbb{R}^\kappa$, then for no additional cost, we may in fact compute a sign invariant semialgebraic decomposition of $\mathbb{R}^\kappa$ for all the sign classes of $\mathcal{F}$, using the procedure described above. Such a decomposition leads to a semialgebraic cell-complex, called *cylindrical algebraic decomposition* (CAD). This notion will be made more precise below. Note that since we have an algebraic sample point for each cell, we can compute the sign assignment with respect to $\mathcal{F}$ of each cell of the decomposition and hence determine exactly those cells whose union constitutes S .

8.6.5 Cylindrical Algebraic Decomposition

Definition 8.6.8 [Cylindrical Algebraic Decomposition (CAD)] A *cylindrical algebraic decomposition* (CAD) of $\mathbb{R}^{\kappa}$ is defined inductively as follows:

- **BASE CASE:** $n = 1$.
A partition of $\mathbb{R}^{\mathbb{K}}$ into a finite set of algebraic numbers, and into the finite and infinite open intervals bounded by these numbers.
- **INDUCTIVE CASE:** $n > 1$.
Assume inductively that we have a CAD $\mathcal{K}'$ of $\mathbb{R}^{\kappa - \mathbb{K}}$. Define a CAD $\mathcal{K}$ of $\mathbb{R}^{\kappa}$ via an *auxiliary polynomial*

$$g_{C'}(\bar{x}, x_n) = g_{C'}(x_1, \dots, x_{n-1}, x_n) \in \mathbb{Q}[\curvearrowright_{\mathbb{K}}, \dots, \curvearrowright_{\mathbb{K}}],$$

one per each $C' \in \mathcal{K}'$. The cells of $\mathcal{K}$ are of two kinds:

1. For each $C' \in \mathcal{K}'$,

$$C' \times (\mathbb{R} \cup \{\pm\infty\}) = \text{cylindrical over } C'.$$

2. For each cell $C' \in \mathcal{K}'$, the polynomial $g_{C'}(p', x_n)$ has m distinct real roots for each $p' \in C'$:

$$r_1(p'), r_2(p'), \dots, r_m(p'),$$

each r_i being a continuous function of p' . The following sectors and sections are cylindrical over C' :

$$\begin{aligned} C_0^* &= \left\{ \langle p', x_n \rangle : x_n \in [-\infty, r_1(p')) \right\}, \\ C_1 &= \left\{ \langle p', x_n \rangle : x_n \in [r_1(p'), r_1(p')] \right\}, \\ C_1^* &= \left\{ \langle p', x_n \rangle : x_n \in (r_1(p'), r_2(p')) \right\}, \\ C_2 &= \left\{ \langle p', x_n \rangle : x_n \in [r_2(p'), r_2(p')] \right\}, \\ &\vdots \\ C_m &= \left\{ \langle p', x_n \rangle : x_n \in [r_m(p'), r_m(p')] \right\}, \\ C_m^* &= \left\{ \langle p', x_n \rangle : x_n \in (r_m(p'), +\infty] \right\}. \end{aligned}$$

See Figure 8.4 for an example of a cylindrical algebraic decomposition of $\mathbb{R}^{\mathbb{K}}$. $\square$

Let

$$\mathcal{F} = \left\{ f_{i,j}(x_1, \dots, x_n) \in \mathbb{Q}[\curvearrowright_{\mathbb{K}}, \dots, \curvearrowright_{\mathbb{K}}] : \curvearrowright = \mathbb{K}, \dots, \succ, \curvearrowleft = \mathbb{K}, \dots, \prec \right\}.$$

In order to compute a cylindrical algebraic decomposition of $\mathbb{R}^{\kappa}$, which is $\mathcal{F}$ -sign-invariant, we follow the following three steps:

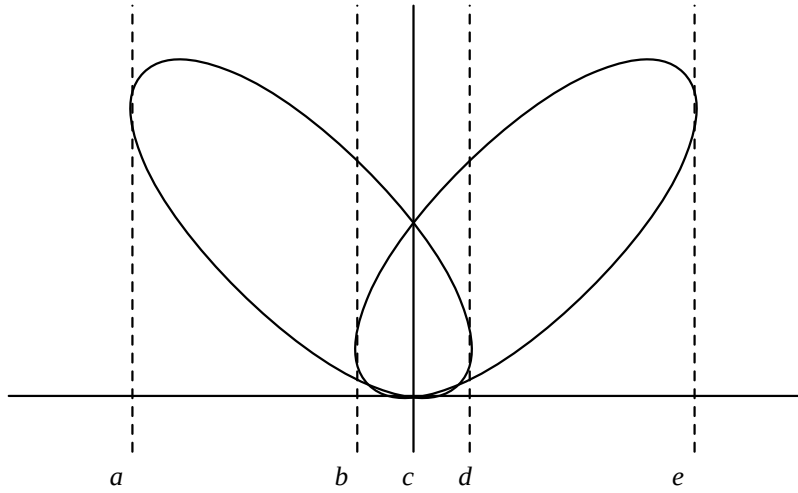


Figure 8.4: Cylindrical algebraic decomposition.

1. **Project:** Compute the $(n-1)$ -variate polynomials $\Phi(\mathcal{F})$. Note that if $|\mathcal{F}|$ is the number of polynomials in $\mathcal{F}$ and d is the maximum degree of any polynomial in $\mathcal{F}$, then

$$|\Phi(\mathcal{F})| = O(d |\mathcal{F}|^2) \quad \text{and} \quad \deg(\Phi(\mathcal{F})) = O(d^2).$$

2. **Recur:** Apply the algorithm recursively to compute a CAD of $\mathbb{R}^{k-1}$ which is $\Phi(\mathcal{F})$ -sign-invariant.
3. **Lift:** Lift the $\Phi(\mathcal{F})$ -sign-invariant CAD of $\mathbb{R}^{k-1}$ up to a $\mathcal{F}$ -sign-invariant CAD of $\mathbb{R}^k$ using the auxiliary polynomial $\Pi(\mathcal{F})$ of degree no larger than $d |\mathcal{F}|$.

It is easy to see how to modify the above procedure in order that we also have a sample point (with algebraic number coordinates) for each cell of the final cell decomposition. The complete algorithm is as follows:

CAD($\mathcal{F}$)

Input: $\mathcal{F} \subseteq \mathbb{Q}[\curvearrowright_1, \dots, \curvearrowright_k]$.

Output: A $\mathcal{F}$ -sign-invariant CAD of $\mathbb{R}^k$.

if $n = 1$ then

Decompose $\mathbb{R} \cup \{\pm\infty\}$ by the set of real roots of the polynomials of $\mathcal{F}$; Compute the sample points to be these real roots and their midpoints;

```

elseif  $n > 1$  then
  Construct  $\Phi(\mathcal{F}) \subseteq \mathbb{Q}[\sqrt[n]{x_1}, \dots, x_{n-1}]$ ;

   $\mathcal{K}' := \text{CAD}(\Phi(\mathcal{F}))$ ;
  Comment:  $\mathcal{K}'$  is a  $\Phi(\mathcal{F})$ -sign-invariant CAD of  $\mathbb{R}^{n-\mu}$ .

  for each  $C' \in \mathcal{K}'$  loop
    Construct  $\Pi(\mathcal{F})$ , the product of those polynomials of  $\mathcal{F}$ 
    that do not vanish at some sample point  $\alpha_{C'} \in C'$ ; De-
    compose  $\mathbb{R} \cup \{\pm\infty\}$  by the roots of  $\Pi(\mathcal{F})$  into sections and
    sectors;
    Comment: The decomposition leads to a decomposition
     $\mathcal{K}_{C'}$  of  $C' \times (\mathbb{R} \cup \{\pm\infty\})$ ;

    The sample points above  $C'$ :

    
$$\begin{aligned}
    &\langle \alpha_{C'}, r_1(\alpha_{C'}) - 1 \rangle, \\
    &\langle \alpha_{C'}, r_1(\alpha_{C'}) \rangle, \\
    &\langle \alpha_{C'}, (r_1(\alpha_{C'}), r_2(\alpha_{C'}))/2 \rangle, \\
    &\langle \alpha_{C'}, r_2(\alpha_{C'}) \rangle, \\
    &\vdots \\
    &\langle \alpha_{C'}, r_m(\alpha_{C'}) \rangle, \\
    &\langle \alpha_{C'}, r_m(\alpha_{C'}) + 1 \rangle,
    \end{aligned}$$


    where  $r_i$ 's are the real root functions for  $\Pi(\mathcal{F})$ ;

    Each cell of the  $\mathcal{K}_{C'}$  has a propositional defining sentence
    involving sign sequences for  $\Phi(\mathcal{F})$  and  $\mathcal{F}$ ;
  end{loop };

   $\mathcal{K} := \bigcup_{C' \in \mathcal{K}'} \mathcal{K}_{C'}$ ;
end{if };

return  $\mathcal{K}$ ;
end{CAD}  $\square$ 

```

Complexity

If we assume that the dimension n is a fixed constant, then the algorithm CAD is polynomial in $|\mathcal{F}|$ and $\deg(\mathcal{F})$. However, the algorithm can be easily seen to be double exponential in n as the number of polynomials produced at the lowest dimension is

$$\left(|\mathcal{F}| \deg(\mathcal{F})\right)^{2^{O(n)}},$$

each of degree no larger than $d^{2^{O(n)}}$. Also, the number of cells produced

by the algorithm is given by the *double exponential* function

$$\left(|\mathcal{F}| \deg(\mathcal{F})\right)^{2^{O(n)}},$$

while it is known that the total number of $\mathcal{F}$ -sign-invariant connected components are bounded by the following *single-exponential* function:

$$\left(\frac{O(|\mathcal{F}| \deg(\mathcal{F}))}{n}\right)^n.$$

In summary, we have the following:

Theorem 8.6.8 (Collin's Theorem) *Given a finite set of multivariate polynomials*

$$\mathcal{F} \subseteq \mathbb{Q}[\curvearrowright_{\mathbb{K}}, \dots, \curvearrowright_{\mathbb{K}}],$$

we can effectively construct the followings:

- *An $\mathcal{F}$ -sign-invariant cylindrical algebraic decomposition of $\mathbb{R}^{\times}$ into semialgebraic connected cells. Each cell $C \in \mathcal{K}$ is homeomorphic to $\mathbb{R}^{\delta}$, for some $0 \leq \delta \leq n$.*
- *A sample algebraic point p_C in each cell $C \in \mathcal{K}$ and defining polynomials for each sample point p_C .*
- *Quantifier-free defining sentences for each cell $C \in \mathcal{K}$. $\square$*

Furthermore, the cylindrical algebraic decomposition produced by the CAD algorithm is a *cell complex*, if the set of defining polynomials

$$\mathcal{F} \subseteq \mathbb{Q}[\curvearrowright_{\mathbb{K}}, \dots, \curvearrowright_{\mathbb{K}}],$$

is *well-based* in $\mathbb{R}^{\times}$ in the sense that the following nondegeneracy conditions hold:

1. For all $p' \in \mathbb{R}^{\times - \mathbb{K}}$,

$$\left(\forall f_i \in \mathcal{F}\right) \left[f_i(p', x_n) \neq 0\right].$$

2. $\Phi(\mathcal{F})$ is *well-based* in $\mathbb{R}^{\times - \mathbb{K}}$. That is, For all $p'' \in \mathbb{R}^{\times - \mathbb{K}}$,

$$\left(\forall g_j \in \Phi(\mathcal{F})\right) \left[g_j(p'', x_{n-1}) \neq 0\right],$$

and so on.

The resulting CAD is said to be *well-based*. Also note that, given an $\mathcal{F}$, there is always a linear change of coordinates that results in a well-based system of polynomials. As a matter of fact, any random change of coordinates will result in a well-based system *almost surely*.

Theorem 8.6.9 *If the cellular decomposition $\mathcal{K}$ produced by the CAD algorithm is well-based, then $\mathcal{K}$ is a semialgebraic cell complex.*

PROOF.

As a result of Collin's theorem, we only need to show that

Closure of each cell $C_i \in \mathcal{K}$, $\overline{C_i}$ is a union of some cells C_j 's:

$$\overline{C_i} = \bigcup_j C_j.$$

The proof proceeds by induction on the dimension, n . When $n = 1$, it is easy to see that the decomposition is a cell complex as the decomposition consists of zero-dimensional closed cells (points) or one-dimensional open cells (open intervals) whose limit points (endpoints of the interval) are included in the decomposition.

Let $C_i \in \mathcal{K}$ be a cell in $\mathbb{R}^\times$, which is cylindrical over some cell $C'_k \in \mathcal{K}'$, a CAD of $\mathbb{R}^{\times-\mathbb{K}}$. By the inductive hypothesis, we may assume that $\mathcal{K}'$ is a cell complex and

$$\overline{C'_k} = C'_k \cup C'_{k_1} \cup C'_{k_2} \cup \dots \cup C'_{k_l},$$

where C'_{k_i} 's are in $\mathcal{K}'$.

We show that

1. If C_i is a *section*, then $\overline{C_i}$ consists of
 - (a) C_i itself.
 - (b) Limit points of C_i . These are comprised of sections cylindrical over cells in $\partial C'_k$.
2. If C_i is a *sector*, then $\overline{C_i}$ consists of:
 - (a) C_i itself.
 - (b) Limit points of C_i . These are comprised of upper and lower bounding sections for C_i , cylindrical over C'_K , and sectors and sections cylindrical over cells in $\partial C'_k$.

The key idea is to show that, since sections are given by some continuous real root function $r_j(p')$, the closure of a particular section C_i is simply the image of a real root function over some cell $C'_{k_m} \subseteq \partial C'_k$ which extends $r_j(p')$.

The proof is by contradiction: consider a sequence of points $p'_1, p'_2, p'_3, \dots$, in C'_k , which converges to some point $p'^* \in C'_{k_1} \subseteq \partial C'_k$, say. This sequence of points can be lifted to a sequence of points in the section C_i by the real root function r_j :

$$p_1 = \langle p'_1, r_j(p'_1) \rangle, \quad p_2 = \langle p'_2, r_j(p'_2) \rangle, \quad p_3 = \langle p'_3, r_j(p'_3) \rangle, \quad \dots$$

For every neighborhood N containing p'^* , consider its image under the map r_j . The intersection of all such neighborhoods must be a connected interval of $J \subseteq p'^* \times \mathbb{R}$. Also, all the defining polynomials $\mathcal{F}$ must vanish over J . But as a direct consequence of the well-basedness assumption, we find that J must be a point contained in the image of a real root function over $C'_{k_1} \subseteq \partial C'_k$. The rest follows from a direct examination of the geometry of a cylindrical algebraic decomposition. $\square$

The resulting cell complex is usually represented by a labeled directed graph $G = \langle V, E, \delta, \sigma \rangle$, where

$$\begin{aligned} V &= \text{vertices representing the cells} \\ E &= \text{edges representing the incidence relation among the cells} \\ &\quad uEv \Leftrightarrow C_u \subseteq \overline{C_v} \\ \delta &: V \rightarrow \mathbb{N} = \text{dimension of the cells} \\ \sigma &: V \rightarrow \{-1, 0, +1\} = \text{sign assignment to the cells} \end{aligned}$$

Such a graph allows one to study the connectivity structures of the cylindrical decomposition, and has important applications to robotics path planning. G is said to be a *connectivity graph of a cell complex*.

8.6.6 Tarski Geometry

Tarski sentences are semantic clauses in a first-order language (defined by Tarski in 1930) of equalities, inequalities, and inequations of algebraic functions over the real. Such sentences may be constructed by introducing the following quantifiers, “ $\forall$ ” (universal quantifier) and “ $\exists$ ” (existential quantifier), to the propositional algebraic sentences. The quantifiers are assumed to range over the real numbers.

Let $\mathcal{Q}$ stand for a quantifier (either universal $\forall$ or existential $\exists$). If $\phi(y_1, \dots, y_r)$ is a propositional algebraic sentence, then it is also a first-order algebraic sentence. All The variables y 's are *free* in ϕ . Let $\Phi(y_1, \dots, y_r)$ and $\Psi(z_1, \dots, z_s)$ be two first-order algebraic sentences (with free variables y 's and z 's, respectively); then a sentence combining Φ and Ψ by a Boolean connective is a first-order algebraic sentence with free variables $\{y_i\} \cup \{z_i\}$. Lastly, let $\Phi(y_1, \dots, y_r, x)$ be a first-order algebraic sentence (with free variables x and y), then

$$(\mathcal{Q} x) [\Phi(y_1, \dots, y_r, x)]$$

is a first-order algebraic sentence with only y 's as the free variables. The variable x is *bound* in $(\mathcal{Q} x)[\Phi]$.

A Tarski sentence $\Phi(y_1, \dots, y_r)$ with free variable y 's is said to be true, if for all $\langle \zeta_1, \dots, \zeta_r \rangle \in \mathbb{R}^{\setminus}$

$$\Phi(\zeta_1, \dots, \zeta_r) = \text{True}.$$

Example 8.6.9 1. Let $f(x) \in \mathbb{Z}[\curvearrowright]$ have the following real roots:

$$\alpha_1 < \dots < \alpha_{j-1} < \alpha_j < \dots.$$

Then the algebraic number α_j can be expressed as

$$\begin{aligned} f(y) = 0 \quad \wedge \quad & \left(\exists x_1, \dots, x_{j-1} \right) \\ & \left[(x_2 - x_1 > 0) \wedge \dots \wedge (x_{j-1} - x_{j-2} > 0) \right. \\ & \quad \wedge (f(x_1) = 0) \wedge \dots \wedge (f(x_{j-1}) = 0) \\ & \quad \wedge (\forall z) [(f(z) = 0 \wedge y - z > 0) \\ & \quad \quad \Rightarrow ((z - x_1 = 0) \vee \dots \vee (z - x_{j-1} = 0))] \left. \right]. \end{aligned}$$

If (l, r) is an isolating interval for α_j , we could also express the real root α_j by the following Tarski sentence:

$$\begin{aligned} & (f(y) = 0) \wedge (y - l > 0) \wedge (r - y > 0) \\ & \wedge \left(\forall x \right) \left[((x - y \neq 0) \wedge (x - l > 0) \wedge (r - x > 0)) \Rightarrow f(x) \neq 0 \right]. \end{aligned}$$

2. Consider the following Tarski sentence:

$$\left(\exists x \right) \left(\forall y \right) \left[(y^2 - x > 0) \right].$$

The sentence can be seen to be true, since if we choose a strictly negative number as a value for x , then for all y , the difference of y^2 and x is always strictly positive.

Next, consider the following Tarski sentence:

$$\left(\exists x \right) \left(\forall y \right) \left[(y^2 - x < 0) \right].$$

The sentence can be seen to be false.

3. Let $S \subseteq \mathbb{R}^\times$ be a semialgebraic set; then its closure $\overline{S}$ can be defined by the following Tarski sentence:

$$\Psi(\bar{x}) = \left(\forall \epsilon \right) \left[(\epsilon > 0) \Rightarrow (\exists \bar{y}) [\Phi_S(\bar{y}) \wedge \|\bar{x} - \bar{y}\|_2 < \epsilon] \right],$$

where Φ_S is a defining formula for S . $\overline{S} = \{\bar{x} : \Psi(\bar{x})\}$.

4. Let C_1 and C_2 be two cells of a cylindrical algebraic decomposition of $\mathbb{R}^k$, with defining formulas Φ_{C_1} and Φ_{C_2} , respectively. C_1 and C_2 are *adjacent* if and only if

$$C_1 \cap \overline{C_2} \neq \emptyset \quad \text{or} \quad \overline{C_1} \cap C_2 \neq \emptyset.$$

The following Tarski sentence characterizes the adjacency relation:

$$\begin{aligned} & (\exists \bar{x}) (\forall \epsilon) (\exists \bar{y}) \\ & \left[(((\epsilon > 0) \wedge \Phi_{C_1}(\bar{x})) \Rightarrow (\Phi_{C_2}(\bar{y}) \wedge (\|\bar{x} - \bar{y}\|_2 < \epsilon))) \right. \\ & \left. \vee (((\epsilon > 0) \wedge \Phi_{C_2}(\bar{x})) \Rightarrow (\Phi_{C_1}(\bar{y}) \wedge (\|\bar{x} - \bar{y}\|_2 < \epsilon))) \right]. \end{aligned}$$

Since we shall produce an effective decision procedure for Tarski sentences, we see that one can construct the connectivity graph of a cell complex effectively, provided that we have the defining formulas for the cells of the cell complex. $\square$

A Tarski sentence is said to be *prenex* if it has the form

$$(\mathcal{Q} x_1) (\mathcal{Q} x_2) \cdots (\mathcal{Q} x_n) [\phi(y_1, y_2, \dots, y_r, x_1, \dots, x_n)],$$

where ϕ is quantifier-free. The string of quantifiers $(\mathcal{Q} x_1) (\mathcal{Q} x_2) \cdots (\mathcal{Q} x_n)$ is called the *prefix* and ϕ is called the *matrix*. Given a Tarski sentence Ψ , a prenex Tarski sentence logically equivalent to Ψ is called its *prenex form*.

The following procedure shows that for every Tarski sentence, one can find its prenex form:

1. STEP 1: Eliminate redundant quantifiers. Replace a subformula $(\mathcal{Q} x)[\Phi]$ by Φ , if x does not occur in Φ .
2. STEP 2: Rename variables such that the same variable does not occur as free and bound. If there are two subformulas $\Psi(x)$ and $(\mathcal{Q} x)[\Phi(x)]$ at the same level, replace the latter by $(\mathcal{Q} x_{\text{new}})[\Phi(x_{\text{new}})]$, where x_{new} is a new variable not occurring before.
3. STEP 3: Move negations ($\neg$) inward.

$$\begin{aligned} \neg(\forall x)[\Phi(x)] & \rightarrow (\exists x)[\neg\Phi(x)] \\ \neg(\exists x)[\Phi(x)] & \rightarrow (\forall x)[\neg\Phi(x)] \\ \neg(\Phi \vee \Psi) & \rightarrow (\neg\Phi \wedge \neg\Psi) \\ \neg(\Phi \wedge \Psi) & \rightarrow (\neg\Phi \vee \neg\Psi) \\ \neg\neg\Phi & \rightarrow \Phi \end{aligned}$$

4. STEP 4: Push quantifiers to the left.

$$\begin{aligned} (\mathcal{Q} x)[\Phi(x)] \wedge \Psi &\rightarrow (\mathcal{Q} x)[\Phi(x) \wedge \Psi] \\ (\mathcal{Q} x)[\Phi(x)] \vee \Psi &\rightarrow (\mathcal{Q} x)[\Phi(x) \vee \Psi] \\ \Psi \wedge (\mathcal{Q} x)[\Phi(x)] &\rightarrow (\mathcal{Q} x)[\Psi \wedge \Phi(x)] \\ \Psi \vee (\mathcal{Q} x)[\Phi(x)] &\rightarrow (\mathcal{Q} x)[\Psi \vee \Phi(x)] \end{aligned}$$

Example 8.6.10 Consider the following Tarski sentence:

$$(\forall x) \left[((\forall y) [f(x) = 0] \vee (\forall z) [g(z, y) > 0]) \Rightarrow \neg(\forall y) [h(x, y) \leq 0] \right].$$

After eliminating redundant quantifiers and renaming variables,

$$(\forall x) \left[([f(x) = 0] \vee (\forall z) [g(z, y) > 0]) \Rightarrow \neg(\forall w) [h(x, w) \leq 0] \right].$$

After simplification, we have

$$(\forall x) \left[([f(x) \neq 0] \wedge \neg(\forall z) [g(z, y) > 0]) \vee \neg(\forall w) [h(x, w) \leq 0] \right].$$

After moving negations inward,

$$(\forall x) \left[([f(x) \neq 0] \wedge (\exists z) [g(z, y) \leq 0]) \vee (\exists w) [h(x, w) > 0] \right].$$

After pushing the quantifiers outward,

$$(\forall x) (\exists z) (\exists w) \left[((f(x) \neq 0) \wedge (g(z, y) \leq 0)) \vee (h(x, w) > 0) \right]. \quad \square$$

Finally, we are ready to consider an effective procedure to decide whether a given Tarski sentence $\Psi(x_1, \dots, x_r)$ is true. Here $x_1, \dots, x_r$ are assumed to be its free variables, and the polynomials occurring in Ψ are assumed to have rational coefficients.

As a result of our earlier discussion, we may assume that our Tarski sentence is presented in its prenex form and that it is universally closed with respect to its free variables.

$$\begin{aligned} &(\forall x_1) \cdots (\forall x_r) \Psi(x_1, \dots, x_r) \\ &= (\forall x_1) \cdots (\forall x_r) (\mathcal{Q}x_{r+1}) \cdots (\mathcal{Q}x_n) [\psi(x_1, \dots, x_r, x_{r+1}, \dots, x_n)], \end{aligned}$$

ψ is a quantifier-free matrix. Thus from now on we deal only with prenex Tarski sentences with no free variable and where the variables are so ordered that in the prefix the n variables appear in the order

$$x_1, x_2, \dots, x_n.$$

One can describe the decision procedure for such a Tarski sentence in terms of a Player-Adversary game. We start with the following illustrations:

Example 8.6.11 1. The game for the following Tarski sentence proceeds as shown below:

$$(\exists x) (\forall y) [(y^2 - x > 0)].$$

The first quantifier is $\exists$ and the first move is Player's. Player chooses a strictly negative number for x , say -1 . The second quantifier is a $\forall$ and the next move is Adversary's. Now independent of what Adversary chooses for y , we see that

$$y^2 - (-1) = y^2 + 1 > 0.$$

The matrix is true, and hence Player *wins* and the Tarski sentence is true.

2. Next consider the game for the following Tarski sentence:

$$(\exists x) (\forall y) [(y^2 - x < 0)].$$

Again the first quantifier is $\exists$ and the first move is Player's. Let Player choose $x = a$. The second quantifier is a $\forall$ and the next move is Adversary's. Adversary chooses $y = a + (1/2)$, and we have

$$y^2 - x = (a + 1/2)^2 - a = a^2 + 1/4 > 0.$$

The matrix is false, and hence Adversary *wins* and the Tarski sentence is false.

Thus given a Tarski sentence

$$(\mathcal{Q}_1 x_1) \cdots (\mathcal{Q}_n x_n) [\psi(x_1, \dots, x_n)],$$

the i^{th} ($i = 1, \dots, n$) move is as follows: Assume the values selected up to this point are

$$\zeta_1, \zeta_2, \dots, \zeta_{i-1}$$

- If the i^{th} quantifier $\mathcal{Q}_i$ is $\exists$, then it is Player's move; otherwise, if $\mathcal{Q}_i$ is $\forall$, then it is Adversary's move.
- If it is Player's move, he selects $x_i = \zeta_i$ in order to force a win for himself, i.e., he tries to make the following hold:

$$(\mathcal{Q}_{i+1} x_{i+1}) \cdots (\mathcal{Q}_n x_n) [\psi(\zeta_1, \dots, \zeta_i, x_{i+1}, \dots, x_n)] = \text{True}.$$

If it is Adversary's move, he selects $x_i = \zeta_i$ in order to force a win for himself, i.e., he tries to make the following hold:

$$(\mathcal{Q}_{i+1} x_{i+1}) \cdots (\mathcal{Q}_n x_n) [\psi(\zeta_1, \dots, \zeta_i, x_{i+1}, \dots, x_n)] = \text{False}.$$

After all ζ_i 's have been chosen, we evaluate

$$\psi(\zeta_1, \dots, \zeta_n);$$

If it is true, then Player wins (i.e., the Tarski sentence is true); otherwise, Adversary wins (i.e., the Tarski sentence is false). Thus, the sequence of choices in this game results in a point

$$p = \langle \zeta_1, \dots, \zeta_n \rangle \in \mathbb{R}^\kappa,$$

and the final outcome depends on this point p .

Let $\mathcal{F} \subseteq \mathbb{Q}[\curvearrowright, \dots, x_n]$ be the set of polynomials appearing in the matrix ψ . Now consider a cylindrical algebraic decomposition $\mathcal{K}$ of $\mathbb{R}^\kappa$ for $\mathcal{F}$. Let $C_p \in \mathcal{K}$ be the cell containing p . If $q \in C_p$ is a sample point in the cell C_p

$$q = \langle \alpha_1, \dots, \alpha_n \rangle \in \mathbb{R}^\kappa,$$

then the α_i 's constitute a winning strategy for Player (respectively, Adversary) if and only if ζ_i 's also constitute a winning strategy for Player (respectively, Adversary). Thus, *the search could have been conducted only over the coordinates of the sample points in the cylindrical algebraic decomposition*. This leads to an effective procedure, once a cylindrical algebraic decomposition for $\mathcal{F}$, endowed with the sample points, have been computed.

Since the cylindrical algebraic decomposition produces a sequence of decompositions:

$$\mathcal{K}_1 \text{ of } \mathbb{R}^{\kappa_1}, \mathcal{K}_2 \text{ of } \mathbb{R}^{\kappa_2}, \dots, \mathcal{K}_\kappa \text{ of } \mathbb{R}^\kappa,$$

such that the each cell $C_{i-1,j}$ of $\mathcal{K}_i$ is cylindrical over some cell C_{i-1} of $\mathcal{K}_{i-1}$, the search progresses by first finding cells C_1 of $\mathcal{K}_1$ such that

$$(\mathcal{Q}_2 x_2) \cdots (\mathcal{Q}_n x_n) [\psi(\alpha_{C_1}, x_2, \dots, x_n)] = \text{True}.$$

For each C_1 , the search continues over cells C_{12} of $\mathcal{K}_2$ cylindrical over C_1 such that

$$(\mathcal{Q}_3 x_3) \cdots (\mathcal{Q}_n x_n) [\psi(\alpha_{C_1}, \alpha_{C_{12}}, x_3, \dots, x_n)] = \text{True},$$

etc. Finally, at the bottom level the truth properties of the matrix ψ are evaluated at all the sample points.

This produces a tree structure, where each node at the $(i-1)^{\text{th}}$ level corresponds to a cell $C_{i-1} \in \mathcal{K}_{i-1}$ and its children correspond to the cells $C_{i-1,j} \in \mathcal{K}_i$ that are cylindrical over C_{i-1} . The leaves of the tree correspond to the cells of the final decomposition $\mathcal{K} = \mathcal{K}_\kappa$. Using the game-theoretic nature of the problem discussed earlier we can further label every node at the $(i-1)^{\text{th}}$ level “AND” (respectively, “OR”) if $\mathcal{Q}_i$ is a universal quantifier $\forall$ (respectively, $\exists$). Such a tree is a so-called AND-OR tree.

The tree may be evaluated as follows: First label the leaves true or false, depending on whether the matrix ψ evaluates to true or false in the corresponding cell. Note that the truth value of ψ depends only on the sign assignment of the cell. Inductively, assuming all the nodes up to level $(i - 1)$ have been labeled, an i^{th} level node is labeled true if it is an AND (respectively, OR) node and all (respectively, some) of its children are labeled true. Finally, the Tarski sentence is true if and only if the root of the tree is labeled true. This constitutes a decision procedure for the Tarski sentences.

Consider a sample point

$$p = \langle \alpha_1, \alpha_2, \dots, \alpha_n \rangle,$$

in some cell $C \in \mathcal{K}$. Assume that the algebraic number α_i has an interval representation $\langle f_i, l_i, r_i \rangle$. The truth value at a leaf corresponding to C , (i.e., $\psi(\alpha_1, \dots, \alpha_n)$) can be expressed by the following logically equivalent quantifier-free sentences involving only polynomials with rational coefficients:

$$\begin{aligned} (f_1(z_1) = f_2(z_2) = \dots = f_n(z_n) = 0) \\ \wedge (l_1 < z_1 < r_1) \wedge \dots \wedge (l_n < z_n < r_n) \\ \wedge \psi(z_1, z_2, \dots, z_n). \end{aligned}$$

Thus each leaf of the AND-OR tree can be expressed as a quantifier-free sentence as above. Now, the tree itself can be expressed as a quantifier-free sentence involving conjunctions and disjunctions for the AND and OR nodes, respectively. Clearly, all the polynomials involved are over $\mathbb{Q}$.

For examples, consider the sentences

$$(\exists x) (\forall y) [(y^2 - x > 0)]$$

and

$$(\exists x) (\forall y) [(y^2 - x < 0)].$$

The sample points for a CAD of $y^2 - x$ are as follows:

$$(-1, 0), \left\{ \begin{array}{c} (0, 1) \\ (0, 0) \\ (0, -1) \end{array} \right\}, \left\{ \begin{array}{c} (1, 2) \\ (1, 1) \\ (1, 1/2) \\ (1, 0) \\ (1, -1/2) \\ (1, -1) \\ (1, -2) \end{array} \right\}$$

The equivalent quantifier-free sentences are

$$\begin{aligned} & (0 > -1) \\ \vee & (1 > 0) \wedge (0 > 0) \wedge (1 > 0) \\ \vee & (4 > 1) \wedge (1 > 1) \wedge (1/4 > 1) \wedge (0 > 1) \wedge (1/4 > 1) \wedge (1 > 1) \wedge (4 > 1), \end{aligned}$$

and

$$\begin{aligned} & (0 < -1) \\ \vee & (1 < 0) \wedge (0 < 0) \wedge (1 < 0) \\ \vee & (4 < 1) \wedge (1 < 1) \wedge (1/4 < 1) \wedge (0 < 1) \wedge (1/4 < 1) \wedge (1 < 1) \wedge (4 < 1). \end{aligned}$$

By a simple examination we see that the first sentence is true, while the second is false.

In summary, we have the following:

Theorem 8.6.10 *Let Φ be a Tarski sentence involving polynomials with rational coefficients. Then we have the following:*

- *There is an effective decision procedure for Φ .*
- *There is a quantifier-free propositional sentence ϕ logically equivalent to Φ . The sentence ϕ involves only polynomials with rational coefficients. $\square$*

Corollary 8.6.11 *Tarski sets (subsets of $\mathbb{R}^k$ defined by a Tarski sentence) are exactly the semialgebraic sets. $\square$*

Problems

Problem 8.1

In an ordered field K , we define $|x|$ for all $x \in K$ as follows:

$$|x| = \begin{cases} x, & \text{if } x \geq 0; \\ -x, & \text{if } x < 0. \end{cases}$$

Prove that for all $x, y \in K$,

- (i) $|x + y| \leq |x| + |y|$.
- (ii) $|xy| = |x||y|$.

Problem 8.2

Give a proof for the following:

Let K be a real closed field and $f(x) \in K[x]$. If $a, b \in K$, $a < b$, then

$$\left(\exists c \in (a, b) \right) \left[f(b) - f(a) = (b - a)D(f)(c) \right].$$

This is the so-called *intermediate value theorem*.

Problem 8.3

Is it true that every integral polynomial $f(x) \in \mathbb{Z}[\wedge]$ has all its real roots in the closed interval $[-\|f\|_\infty, \|f\|_\infty]$?

Hint: Consider the polynomial $x^2 - x - 1$.

Problem 8.4

Show that an algebraic number has a unique minimal polynomial up to *associativity*.

Problem 8.5

Consider a real univariate polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_mx^m + a_1x + a_0 \in \mathbb{R}[\wedge],$$

and u a real root of f .

(i) *Lagrange-Maclaurin's Inequality*. Let

$$a_{n-1} \geq 0, \dots, a_{m+1} \geq 0, \quad \text{and} \quad a_m < 0.$$

Prove that

$$u < 1 + \left(\min(a_i) \right)^{1/(n-m)}.$$

Hint: Assume that $u > 1$ and

$$\begin{aligned} 0 = f(u) &= u^n + a_{n-1}u^{n-1} + \cdots + a_mu^m + a_1u + a_0 \\ &> u^n - \min(a_i)(u^m + u^{m-1} + \cdots + u + 1) \\ &= u^n - \min(a_i) \left(\frac{u^{m+1} - 1}{u - 1} \right). \end{aligned}$$

(ii) *Cauchy's Inequality*. Let

$$a_{m_1} < 0, a_{m_2} < 0, \dots, a_{m_k} < 0, \quad (m_1 > m_2 > \cdots > m_k),$$

be the only negative coefficients of f . Prove that

$$u \leq \max_j \left((k |a_{m_j}|)^{1/(n-m_j)} \right).$$

Hint: Assume that $u > 1$ and

$$\begin{aligned} 0 = f(u) &= u^n + a_{n-1}u^{n-1} + \cdots + a_mu^m + a_1u + a_0 \\ &> u^n - a_{m_1}u^{m_1} - \cdots - a_{m_k}u^{m_k} \\ &\geq u^n - k \max_j (|a_{m_j}| u^{m_j}). \end{aligned}$$

Problem 8.6

Let $f(x) \in \mathbb{R}[\curvearrowright]$ be a univariate real polynomial of positive degree $n > 0$.

(i) Consider f 's Fourier sequence:

$$\overline{\text{FOURIER}}(f) = \left\langle \begin{array}{l} f^{(0)}(x) = f(x), \\ f^{(1)}(x) = f'(x), \\ f^{(2)}(x), \\ \vdots \\ f^{(n)}(x) \end{array} \right\rangle,$$

where $f^{(i)}$ denotes the i^{th} derivative of f with respect to x . Prove the following:

Budan-Fourier Theorem.

Let $f(x) \in \mathbb{R}[\curvearrowright]$ and a and $b \in \mathbb{R}$ be two real numbers with $a < b$. Then

$$\begin{cases} \# \text{ real roots of } f \text{ (counted with multiplicity) in } (a, b) \\ \leq \text{Var}_a(\overline{\text{FOURIER}}(f)) - \text{Var}_b(\overline{\text{FOURIER}}(f)), \quad \text{and} \\ \equiv \text{Var}_a(\overline{\text{FOURIER}}(f)) - \text{Var}_b(\overline{\text{FOURIER}}(f)) \pmod{2}. \end{cases}$$

(ii) Using the Budan-Fourier theorem, present a proof for the following:

Descartes' Theorem.

Let $f(x) \in \mathbb{R}[\curvearrowright]$ be as follows:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0,$$

and

$$V(f) = \text{Var}(\langle a_n, a_{n-1}, \dots, a_0 \rangle).$$

Then the number of strictly positive real roots of f (counted with multiplicity) does not exceed $V(f)$ and is congruent to $V(f) \pmod{2}$.

Problem 8.7

Let $\langle h_0, h_1, \dots, h_s \rangle$ be a Sturm sequence of f and f' .

Let V be the number of sign variations in the sequence

$$\overline{v} = \langle \text{Hcoef}(h_0), \text{Hcoef}(h_1), \dots, \text{Hcoef}(h_s) \rangle,$$

and W be the number of sign variations in the sequence

$$\overline{w} = \langle \text{SHcoef}(h_0), \text{SHcoef}(h_1), \dots, \text{SHcoef}(h_s) \rangle,$$

where $\text{Hcoef}(h)$ is the leading coefficient of h and

$$\text{SHcoef}(h) = (-1)^{\deg(h)} \text{Hcoef}(h)$$

is the “*sign-adjusted*” leading coefficient of h .

Show that the number of distinct real roots of f is $W - V$.

Compute $\overline{v}$ and $\overline{w}$ for a quadratic polynomial $x^2 + bx + c$. What is $W - V$ as a function of b and c ? Can you derive an algebraic criterion for the number of real zeros of $x^2 + bx + c$?

Problem 8.8

Let f , g_1 and g_2 be a *simple set of polynomials* in $K[x]$ (K = a real closed field), in the sense that all their roots are distinct, i.e., they are all square-free and pairwise relatively prime.

(i) For any interval $[a, b] \subseteq K$ ($a < b$), show that

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_f [g_1 > 0, g_2 > 0]_a^b \\ c_f [g_1 > 0, g_2 < 0]_a^b \\ c_f [g_1 < 0, g_2 > 0]_a^b \\ c_f [g_1 < 0, g_2 < 0]_a^b \end{bmatrix}$$

$$= \begin{bmatrix} \text{Var} \left[\overline{\text{STURM}}(f, f') \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g_1) \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g_2) \right]_a^b \\ \text{Var} \left[\overline{\text{STURM}}(f, f'g_1g_2) \right]_a^b \end{bmatrix}.$$

(ii) Show how the preceding formulation can be generalized to the case when f , g_1 and g_2 are *not simple*.

(iii) Show how to obtain a further generalization of this formulation when there are more than two g 's: $f, g_1, g_2, \dots, g_n$.

(iv) Suppose you are given a system of polynomial inequalities, involving polynomials $g_1, g_2, \dots, g_n$ in $\mathbb{R}[\curvearrowright]$, as follows:

$$\begin{aligned} g_1(x) &\leq 0, \\ g_2(x) &\leq 0, \\ &\vdots \\ g_n(x) &\leq 0, \end{aligned}$$

where the notation " $g_i(x) \leq 0$ " represents one of the following three relations: $g_i(x) < 0$, $g_i(x) = 0$ or $g_i(x) > 0$. Using the formulation (iii) and linear algebra, devise a process to determine if there is a solution x at which all the inequalities are satisfied.

Hint: Construct a sequence $f, g_1, g_2, \dots, g_n$ as in (iii), where f is such that, for any sign assignment to g_i 's, the interval corresponding to this assignment, f has a zero:

$$\begin{aligned} G(x) &= g_1(x) g_2(x) \cdots g_n(x), \\ \tilde{G}(x) &= \frac{G(x)}{\text{GCD}(G(x), G'(x))}. \end{aligned}$$

Let

$$f(x) = \tilde{G}(x) \tilde{G}'(x)(x - N)(x + N),$$

where

$$N = 1 + \max \left| \frac{a_i}{a_k} \right|,$$

the a_i 's are coefficients of $\tilde{G}(x)$, and $a_k = \text{Hcoef}(\tilde{G}(x))$.

Problem 8.9

Let $f(x)$ and $g(x) \in \mathbb{Z}[\curvearrowright]$ be two arbitrary integral polynomials of positive degrees $m = \deg(f)$ and $n = \deg(g)$, respectively. Show that

$$\left(\forall \alpha, \text{ s.t. } g(\alpha) = 0 \right) \left[f(\alpha) = 0 \quad \text{or} \quad |f(\alpha)| > \frac{1}{1 + \|g\|_1^m (1 + \|f\|_1)^n} \right].$$

Hint: Consider the zeros of the resultant, $\text{Resultant}_x(g(x), f(x) - y)$.

Problem 8.10

Consider the following monic irreducible integral polynomial $f(x)$,

$$f(x) = x^n - 2(ax - 1)^2, \quad n \geq 3, a \geq 3, a \in \mathbb{Z}.$$

Show that

$$\text{Separation}(f) < 2 \frac{1}{(\|f\|_1/4)^{n+2/4}}.$$

Hint: Show that $f(1/a) > 0$ and $f(1/a \pm h) < 0$, for $h = a^{-(n+2)/2}$.

Problem 8.11 (*Hadamard-Collins-Horowitz Inequality.*)

If S is a commutative ring, a *seminorm* for S is a function

$$\nu: S \rightarrow \{r \geq 0 : r \in \mathbb{R}\}$$

satisfying the following three conditions: For all $a, b \in S$,

$$\nu(a) = 0 \Leftrightarrow a = 0, \quad (8.1)$$

$$\nu(a+b) \leq \nu(a) + \nu(b), \quad (8.2)$$

$$\nu(ab) \leq \nu(a)\nu(b). \quad (8.3)$$

(i) Show that $\|a\|_1 = |a|$ is a seminorm for the integers $\mathbb{Z}$.

(ii) Show the following:

1. If ν is a seminorm over S , then its extension to $S[x]$ defined below

$$\nu(a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0) = \nu(a_n) + \nu(a_{n-1}) + \cdots + \nu(a_0)$$

is also a seminorm over $S[x]$.

2. If ν is a seminorm over S , then its extension to an arbitrary matrix $M \in S^{m \times n}$

$$\nu(M) = \sum_{i=1}^m \sum_{j=1}^n \nu(M_{i,j})$$

satisfies the conditions (8.1), (8.2) and (8.3), whenever the operations are defined.

(iii) If M is a square matrix over a commutative ring S with a seminorm ν , then show that the following generalization of Hadamard's inequality holds:

$$\nu(\det M) \leq \prod_i \nu(M_i),$$

where M_i is the i^{th} row of M and $\det M$ is the determinant of M .

Problem 8.12

Consider an $n \times n$ polynomial matrix with integral polynomial entries

$$M(x) = \begin{bmatrix} A_{1,1}(x) & A_{1,2}(x) & \cdots & A_{1,n}(x) \\ A_{2,1}(x) & A_{2,2}(x) & \cdots & A_{2,n}(x) \\ \vdots & \vdots & \ddots & \vdots \\ A_{n,1}(x) & A_{n,2}(x) & \cdots & A_{n,n}(x) \end{bmatrix}.$$

Prove that

$$\det(M(x)) \equiv 0 \quad \Leftrightarrow \quad \det\left(M\left(\frac{1}{2(adn)^n}\right)\right) = 0,$$

where

$$a = \max_{i,j} \|A_{i,j}\|_{\infty} \quad \text{and} \quad d = \max_{i,j} \deg(A_{i,j}).$$

Problem 8.13

Consider two real algebraic numbers α and β defined by two polynomials $f(x)$ and $g(x)$ of respective positive degrees m and n . Prove that if $\alpha \neq \beta$ then

$$\Delta = |\alpha - \beta| > \frac{1}{2^{(n+1)(m+1)} \|f\|_1^n \|g\|_1^m}.$$

Problem 8.14

Let $f(x) \in \mathbb{C}[\curvearrowright]$ be an arbitrary polynomial:

$$f(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_0.$$

Consider the usual norm for complex numbers:

$$|a + ib| = (a^2 + b^2)^{1/2}.$$

The 1-norm of f can now defined to be

$$\|f\|_1 = |c_n| + |c_{n-1}| + \cdots + |c_0|.$$

Let the zeros of f be enumerated as

$$|\xi_1| \leq |\xi_2| \leq \cdots |\xi_m| \leq 1 < |\xi_{m+1}| \leq \cdots \leq |\xi_n|,$$

counting each zero as many times as its multiplicity.

Let $\mathcal{M}(f)$ be defined as

$$\mathcal{M}(f) = \left| c_n \prod_{i=1}^n \max(1, \xi_i) \right| = |c_n| \prod_{i=1}^n \max(1, |\xi_i|).$$

First we want to derive the following relations:

$$\mathcal{M}(f) \leq \|f\|_1 \leq 2^n \mathcal{M}(f).$$

(i) Prove that

$$\|f\|_1 = \sum |c_n \xi_{i_1} \xi_{i_2} \cdots \xi_{i_l}| \leq 2^n \mathcal{M}(f).$$

(ii) Next show that

$$\log \mathcal{M}(f) = \frac{1}{2\pi} \int_0^{2\pi} \log |f(e^{it})| dt \leq \max_{1 \leq t \leq 2\pi} \log |f(e^{it})| \leq \log \|f\|_1,$$

Thus, concluding that

$$\mathcal{M}(f) \leq \|f\|_1.$$

Hint: Use Jensen's integral formula:

$$\frac{1}{2\pi} \int_0^{2\pi} \log |F(\rho e^{it})| dt = \log |F(0)| + \sum_{i=1}^m \log \frac{\rho}{|\xi_i|}, \quad F(0) \neq 0,$$

where $F(x)$ is a function of the complex variable, regular on the circle of radius ρ . The zeros of F in $|x| \leq \rho$ are given as $\xi_1, \dots, \xi_m$.

(iii) Using these inequalities show that, if f is factorized as follows

$$f(x) = f_1(x) f_2(x) \cdots f_s(x),$$

then

$$\|f_1\|_1 \|f_2\|_1 \cdots \|f_s\|_1 \leq 2^{\deg(f)} \|f\|_1.$$

Problem 8.15

Let $f(x)$ be a polynomial with zeros in $\mathbb{C}$. Then we denote its complex root separation by $\Delta(f)$:

$$\Delta(f) = \min\{|\alpha - \beta| : \alpha \neq \beta \in \mathbb{C} \wedge \mathcal{U}(\alpha) = \mathcal{U}(\beta) = \emptyset\}.$$

Let $f(x)$ be an integral polynomial:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0.$$

Prove the following:

(i) If $f(x)$ is square-free, then

$$\Delta(f) > \sqrt{3} n^{-(n+2)/2} \|f\|_1^{-(n-1)}.$$

(ii) In general,

$$\Delta(f) > \sqrt{3} 2^{-n(n-1)} n^{-(n+2)/2} \|f\|_1^{-(n-1)}.$$

Problem 8.16

(i) Devise a simple and efficient ($O(n^3 \lg n)$ -time) algorithm to convert a real algebraic number from its order representation to its interval representation.

(ii) Prove the following corollary of Thom's lemma:

Consider two real roots ξ and ζ of a real univariate polynomial $f(x) \in \mathbb{R}[\wedge]$ of positive degree $n > 0$. Then

$$\xi > \zeta$$

if and only if, for some $0 \leq m < n$, the following conditions hold:

$$\begin{aligned} \operatorname{sgn}(f^{(m)}(\xi)) &\neq \operatorname{sgn}(f^{(m)}(\zeta)) \\ \operatorname{sgn}(f^{(m+1)}(\xi)) &= \operatorname{sgn}(f^{(m+1)}(\zeta)), \\ &\vdots \\ \operatorname{sgn}(f^{(n)}(\xi)) &= \operatorname{sgn}(f^{(n)}(\zeta)). \end{aligned}$$

and

1. either $\operatorname{sgn}(f^{(m+1)}(\xi)) = +1$ and $f^{(m)}(\xi) > f^{(m)}(\zeta)$,
2. or $\operatorname{sgn}(f^{(m+1)}(\xi)) = -1$ and $f^{(m)}(\xi) < f^{(m)}(\zeta)$.

(iii) Using the corollary above, devise an efficient ($O(n^3 \lg^2 n)$ -time) algorithm to convert a real algebraic number from its sign representation to its interval representation.

Problem 8.17

Prove that if S is a semialgebraic set then its interior, $\operatorname{int}(S)$, closure $\overline{S}$, and boundary $\partial(S) = \overline{S} \setminus \operatorname{int}(S)$ are all semialgebraic.

Problem 8.18

Show that every semialgebraic set is locally connected.

Problem 8.19

Let $S \subseteq \mathbb{R}^\kappa$ be a semialgebraic set. Prove that for some $m \in \mathbb{N}$, there is a *real algebraic set* $T \subseteq \mathbb{R}^{\kappa+\triangleright}$ such that

$$\pi(T) = S,$$

where

$$\begin{aligned} \pi &: \mathbb{R}^{n+m} \rightarrow \mathbb{R}^\kappa \\ &: \langle \xi_1, \dots, \xi_n, \xi_{n+1}, \dots, \xi_{n+m} \rangle \mapsto \langle \xi_1, \dots, \xi_n \rangle \end{aligned}$$

is a natural projection map.

Thus, show that semialgebraic sets constitute the smallest class of subsets of $\mathbb{R}^\kappa$ closed under projection and containing real algebraic sets.

Hint: Let $S \subseteq \mathbb{R}^\kappa$ be defined as follows:

$$S = \left\{ \langle x_1, \dots, x_n \rangle \in \mathbb{R}^\kappa : \text{sgn}(\bigcup_{\mathbb{K}} (\cap_{\mathbb{K}})) = \sim \bigcup \right\},$$

where $i = 1, \dots, m$. Let us now define T as follows:

$$T = \left\{ \langle x_1, \dots, x_n, x_{n+1}, \dots, x_{n+m} \rangle \in \mathbb{R}^\kappa : \sum_{j; s_j < 0} (x_{n+j}^2 f_j + 1)^2 + \sum_{j; s_j = 0} f_j^2 + \sum_{j; s_j > 0} (x_{n+j}^2 f_j - 1)^2 = 0 \right\}.$$

Verify that $S = \pi(T)$.

Problem 8.20

A *robotic system* $\mathcal{R}$ is defined to be a finite collection of rigid compact subparts

$$\{B_1, B_2, \dots, B_m\},$$

where each subpart is assumed to be defined by a piecewise algebraic surface.

A configuration is an n -tuple of parameters that describes the positions and the orientations of the subparts uniquely; the corresponding space of parameters $\mathbb{R}^\kappa$ is called a *configuration space*.

Additionally, we may assume that between every pair of subparts B_i and B_j at most one of the following *holonomic kinematic constraints* may exist:

- **REVOLUTE JOINT:** There is a fixed axis L through a pair of points $p_i \in B_i$ and $p_j \in B_j$ such that B_i and B_j are only allowed to rotate about L .
- **PRISMATIC JOINT:** There is a fixed axis L through a pair of points $p_i \in B_i$ and $p_j \in B_j$ such that B_i and B_j are only allowed to translate about L .

(i) B_i 's are assumed to be able to take any configuration subject to the kinematic constraints such that no two subpart occupies the same space.

A point of the configuration space corresponding to such a configuration is said to be *free*. Otherwise, it is called *forbidden*. The collection of points of configuration space that are free are said to constitute the *free space*, and its complement *forbidden space*.

Show that the *free* and *forbidden spaces* are semialgebraic subsets of $\mathbb{R}^\kappa$.

(ii) Given an *initial* and a desired *final* configurations of the *robotic system*, $\mathcal{R}$, the *motion planning problem* is to decide whether there is a

continuous motion of the subparts from the initial to the final configuration that avoids collision and respects the kinematic constraints.

Devise an algorithm to solve the motion planning problem.

Hint: First compute the connectivity graph for the polynomials defining the free space, and show that there is a continuous motion from one configuration in a free cell to another configuration in a free cell, if there is a path between the vertices corresponding to the respective free cells.

Problem 8.21

Consider the following set of $N = 2^{2^n}$ complex numbers corresponding to the N^{th} roots of unity:

$$\left\{ \alpha_i + i\beta_i : i = 1, \dots, N \right\} \subseteq \mathbb{C}.$$

Define

$$S_n = \left\{ \langle \alpha_i, \beta_i \rangle : i = 1, \dots, N \right\} \subseteq \mathbb{R}^2.$$

Show that there is a Tarski sentence $\Psi_n(x, y)$ with two free variables (x and y) and $O(n)$ quantifiers, $O(n)$ variables, $O(n)$ real linear polynomials, and $O(1)$ real quadratic polynomial such that

$$\left\{ \langle \alpha, \beta \rangle : \Psi_n(\alpha, \beta) = \text{TRUE} \right\} = S_n.$$

Problem 8.22

(i.a) Let $g_i(x, y)$ and $g_j(x, y) \in \mathbb{R}[\wedge, \vee]$. Define

$$\begin{aligned} D(g_i) &= \text{isolated points of } g_i = 0; \\ D(g_i, g_j) &= \text{isolated points of } g_i = 0 \text{ and } g_j = 0 \\ &\quad \setminus (D(g_i) \cup D(g_j)). \end{aligned}$$

If $g_i(x, y)$ and $g_j(x, y)$ are irreducible polynomials, show that

$$|D(g_i)| \leq (\deg(g_i))^2 \quad \text{and} \quad |D(g_i, g_j)| \leq 2 \deg(g_i) \deg(g_j).$$

(i.b) Given a quantifier-free sentence in two variables x and y , involving polynomials

$$\mathcal{F} = \left\{ f_1(x, y), f_2(x, y), \dots, f_m(x, y) \right\} \subseteq \mathbb{R}[\wedge, \vee],$$

show that every isolated point is either in some $D(g_i)$ or some $D(g_i, g_j)$, where g_i 's and g_j 's are irreducible factors of f_i 's.

(i.c) Show that the total number of isolated points of a quantifier-free sentence defined by $\mathcal{F}$ is bounded from above by

$$\left(\sum_{i=1}^m \deg(f_i) \right)^2.$$

Problem 8.23

As direct consequences of the preceding two problems, show the following:

(i) For every $n \in \mathbb{N}$, there exists a quantified Tarski sentence Ψ_n with n quantifiers, of length $O(n)$, and degree $O(1)$ such that any quantifier-free sentence ψ_n logically equivalent to Ψ_n must involve polynomials of

$$\text{degree} = 2^{2^{\Omega(n)}} \quad \text{and} \quad \text{length} = 2^{2^{\Omega(n)}}.$$

(ii) For every $n \in \mathbb{N}$, there exists a quantified Tarski sentence Ψ_n with n quantifiers, of length $O(n)$, and degree $O(1)$ such that Ψ_n induces a cylindrical decomposition, $\mathcal{K}_n$, of $\mathbb{R}^>$ ($m = O(n)$) with

$$|\mathcal{K}_n| = 2^{2^{\Omega(n)}} \text{ cells.}$$

(iii) Thus, argue that both *quantifier elimination problem* and *cylindrical algebraic decomposition problem* have double exponential lower bounds for their time complexity as a function of the input size.

Solutions to Selected Problems

Problem 8.2

First, we choose a linear function $\bar{f}(x)$ such that

$$\tilde{f}(x) = f(x) - \bar{f}(x),$$

vanishes at the points a and b . Hence,

$$\bar{f}(a) = f(a) \quad \text{and} \quad \bar{f}(b) = f(b).$$

Thus,

$$D(\bar{f}) = \frac{f(b) - f(a)}{b - a},$$

and

$$\bar{f}(x) = \frac{f(b) - f(a)}{b - a}x + \frac{bf(a)}{b - a} - \frac{af(b)}{b - a}.$$

Now we can apply Rolle's theorem to $\tilde{f}(x)$ as $\tilde{f}(a) = \tilde{f}(b) = 0$:

$$\left(\exists c \in (a, b) \right) \left[D(\tilde{f})(c) = D(f)(c) - D(\bar{f})(c) = 0 \right].$$

But since, $D(\bar{f}) = (f(b) - f(a))/(b - a)$, we have the necessary conclusion.

We may rewrite the formula above in the following form:

$$f(b) = f(a) + f'(c)(b - a), \quad \text{for some } c \in (a, b).$$

Problem 8.4

Consider an algebraic number α . Let $f(x)$ and $g(x) \in \mathbb{Z}[\sqrt{\cdot}]$ be two nonzero minimal polynomials of α . Clearly $f(x)$ and $g(x)$ have the same degree. Thus, if we consider the following polynomial

$$r(x) = \text{Hcoef}(g) f(x) - \text{Hcoef}(f) g(x) \in \mathbb{Z}[\sqrt{\cdot}],$$

then $\deg(r) < \deg(f) = \deg(g)$ and $r(\alpha) = \text{Hcoef}(g) f(\alpha) - \text{Hcoef}(f) g(\alpha) = 0$. Thus $r(x)$ must be identically zero as, otherwise, it would contradict our assumption that $f(x)$ and $g(x)$ were two minimal polynomials of α . Thus

$$\text{Hcoef}(g) f(x) = \text{Hcoef}(f) g(x) \quad \text{and} \quad f(x) \approx g(x).$$

Problem 8.6

(i) Using Taylor's expansion theorem, we may express the values of the Fourier sequence in a neighborhood of a real number c as follows:

$$\begin{aligned} f(c + \epsilon) &= f(c) + \epsilon f^{(1)}(c) + \frac{\epsilon^2}{2!} f^{(2)}(c) + \cdots + \frac{\epsilon^m}{m!} f^{(m)}(c) \\ &\quad + \cdots + \frac{\epsilon^n}{n!} f^{(n)}(c), \\ f^{(1)}(c + \epsilon) &= f^{(1)}(c) + \epsilon f^{(2)}(c) + \frac{\epsilon^2}{2!} f^{(3)}(c) + \cdots + \frac{\epsilon^{m-1}}{(m+1)!} f^{(m)}(c) \\ &\quad + \cdots + \frac{\epsilon^{n-1}}{(n-1)!} f^{(n)}(c), \\ &\vdots \\ f^{(m)}(c + \epsilon) &= f^{(m)}(c) + \epsilon f^{(m+1)}(c) + \cdots + \frac{\epsilon^{n-m}}{(n-m)!} f^{(n)}(c), \\ &\vdots \\ f^{(n)}(c + \epsilon) &= f^{(n)}(c). \end{aligned}$$

Let us now consider an interval $[a, b]$ containing *exactly one real root* c of f with multiplicity m . Note that it suffices to consider the sign variations in some interval $(c - \epsilon, c + \epsilon) \subseteq [a, b]$. Clearly:

$$f^{(0)}(c) = f^{(1)}(c) = \cdots = f^{(m-1)}(c) = 0 \quad \text{and} \quad f^{(m)}(c) \neq 0.$$

Let $s = \operatorname{sgn}(f^{(m)}(c))$. Hence

$$\begin{array}{lll}
 \operatorname{sgn}(f(c - \epsilon)) & = & (-1)^m s, \quad s = \operatorname{sgn}(f(c + \epsilon)), \\
 \operatorname{sgn}(f^{(1)}(c - \epsilon)) & = & (-1)^{m-1} s, \quad s = \operatorname{sgn}(f^{(1)}(c + \epsilon)), \\
 & \vdots & \vdots \\
 \operatorname{sgn}(f^{(m)}(c - \epsilon)) & = & s = \operatorname{sgn}(f^{(m)}(c + \epsilon)), \\
 \operatorname{sgn}(f^{(m+1)}(c - \epsilon)) & = & \operatorname{sgn}(f^{(m+1)}(c)) = \operatorname{sgn}(f^{(m+1)}(c + \epsilon)), \\
 & \vdots & \vdots \\
 \operatorname{sgn}(f^{(n)}(c - \epsilon)) & = & \operatorname{sgn}(f^{(n)}(c)) = \operatorname{sgn}(f^{(n)}(c + \epsilon)).
 \end{array}$$

Thus it is easily seen that the sequence at the left has exactly m more sign variations than the sequence at the right. Hence, in this case,

$$\begin{aligned}
 & \# \text{ real roots of } f \text{ (counted with multiplicity) in } (a, b) \\
 & = \operatorname{Var}_a(\overline{\operatorname{FOURIER}}(f)) - \operatorname{Var}_b(\overline{\operatorname{FOURIER}}(f)),
 \end{aligned}$$

Next let us consider an interval $[a, b]$ containing *exactly one real root* c of $f^{(k)}$ ($k > 0$) with multiplicity l . Let $m = k + l$. As before, it suffices to consider the sign variations in some interval $(c - \epsilon, c + \epsilon) \subseteq [a, b]$. Clearly:

$$f^{(k)}(c) = f^{(k+1)}(c) = \dots = f^{(m-1)}(c) = 0 \quad \text{and} \quad f^{(m)}(c) \neq 0.$$

Let $s = \operatorname{sgn}(f^{(m)}(c))$. Hence

$$\begin{array}{lll}
 \operatorname{sgn}(f(c - \epsilon)) & = & \operatorname{sgn}(f(c)) = \operatorname{sgn}(f(c + \epsilon)), \\
 & \vdots & \vdots \\
 \operatorname{sgn}(f^{(k-1)}(c - \epsilon)) & = & \operatorname{sgn}(f^{(k-1)}(c)) = \operatorname{sgn}(f^{(1)}(c + \epsilon)), \\
 \operatorname{sgn}(f^{(k)}(c - \epsilon)) & = & (-1)^l s, \quad s = \operatorname{sgn}(f^{(k)}(c + \epsilon)), \\
 \operatorname{sgn}(f^{(k+1)}(c - \epsilon)) & = & (-1)^{l-1} s, \quad s = \operatorname{sgn}(f^{(k+1)}(c + \epsilon)), \\
 & \vdots & \vdots \\
 \operatorname{sgn}(f^{(m)}(c - \epsilon)) & = & s = \operatorname{sgn}(f^{(m)}(c + \epsilon)), \\
 \operatorname{sgn}(f^{(m+1)}(c - \epsilon)) & = & \operatorname{sgn}(f^{(m+1)}(c)) = \operatorname{sgn}(f^{(m+1)}(c + \epsilon)), \\
 & \vdots & \vdots \\
 \operatorname{sgn}(f^{(n)}(c - \epsilon)) & = & \operatorname{sgn}(f^{(n)}(c)) = \operatorname{sgn}(f^{(n)}(c + \epsilon)).
 \end{array}$$

We only need to consider the sign variations from $c - \epsilon$ to $c + \epsilon$ for the following subsequence:

$$\langle f^{(k-1)}, f^{(k)}, \dots, f^{(m)}, f^{(m+1)} \rangle.$$

There are two cases to consider:

1. $\text{sgn}(f^{(k-1)}(c)) = \text{sgn}(f^{(m+1)}(c))$, in which case the subsequence has *even* number of sign variations both at $c - \epsilon$ and $c + \epsilon$, and
2. $\text{sgn}(f^{(k-1)}(c)) \neq \text{sgn}(f^{(m+1)}(c))$, in which case the subsequence has *odd* number of sign variations both at $c - \epsilon$ and $c + \epsilon$;

and in either case the subsequence at the left has more sign variations than the subsequence at the right. Thus,

$$0 \begin{cases} \leq \text{Var}_a(\overline{\text{FOURIER}}(f)) - \text{Var}_b(\overline{\text{FOURIER}}(f)) & \text{and} \\ \equiv \text{Var}_a(\overline{\text{FOURIER}}(f)) - \text{Var}_b(\overline{\text{FOURIER}}(f)) \pmod{2}. \end{cases}$$

(ii) Using Budan-Fourier theorem, we have

$$\begin{aligned} & \# \text{ strictly positive real roots of } f \text{ (counted with multiplicity)} \\ & \begin{cases} \leq \text{Var}_0(\overline{\text{FOURIER}}(f)) - \text{Var}_\infty(\overline{\text{FOURIER}}(f)) & \text{and} \\ \equiv \text{Var}_0(\overline{\text{FOURIER}}(f)) - \text{Var}_\infty(\overline{\text{FOURIER}}(f)) \pmod{2}. \end{cases} \end{aligned}$$

Thus Descartes' theorem follows, once we show that

$$\text{Var}_0(\overline{\text{FOURIER}}(f)) = V(f) \quad \text{and} \quad \text{Var}_\infty(\overline{\text{FOURIER}}(f)) = 0.$$

This is obvious as

$$\text{sgn}_0(\overline{\text{FOURIER}}(f)) = \text{sgn}(\langle a_0, a_1, 2a_2, \dots, m!a_m, \dots, n!a_n \rangle),$$

and

$$\text{sgn}_\infty(\overline{\text{FOURIER}}(f)) = \text{sgn}(\langle a_n, a_n, a_n, \dots, a_n, \dots, a_n \rangle).$$

Problem 8.10

First note that

$$\|f\|_1 = 2a^2 + 4a + 3 \leq (2 + 4/3 + 1/3)a^2, \quad \text{for all } a \geq 3.$$

Hence $a \geq \sqrt{\|f\|_1}/2$ and it suffices to show that

$$\text{Separation}(f) < 2h = 2a^{-(n+2)/2}.$$

Next observe that since

$$\begin{aligned} f(x) &= x^n - 2a^2x^2 + 4ax - 2, \quad \text{and} \\ f(-x) &= (-1)^n x^n - 2a^2x^2 - 4ax - 2, \end{aligned}$$

by Descartes' rule, f has at most two positive real roots and at most one negative real root.

Clearly,

$$f(1/a) = (1/a)^n > 0.$$

Now, notice that

$$f(1/a \pm h) = (1/a \pm h)^n - 2(a(1/a \pm h) - 1)^2 = (1/a \pm h)^n - 2a^2h^2.$$

Thus, choosing $h = a^{-(n+2)/2}$, we have

$$(1/a \pm h)^n \leq (a^{-1} + a^{-(n+2)/2})^n = (1 + a^{-n/2})^n a^{-n} < \left(\frac{413}{243}\right) a^{-n},$$

and

$$f(1/a \pm h) < \left(\frac{413}{243}\right) a^{-n} - 2a^2 a^{-(n+2)} < 0.$$

Thus, f has two positive real roots in the open interval $(1/a - h, 1/a + h)$ and

$$\text{Separation}(f) < \min(1/a - h, 2h) = 2h = 2a^{-(n+2)/2}.$$

Problem 8.11

(i) and (ii) are tedious but straightforward.

(iii) The proof is by induction on the order of the matrix n . The case $n = 1$ is trivial. Consider an $n \times n$ matrix M , $n > 1$. Let $M_{i,j}$ denote the $(i, j)^{\text{th}}$ entry of M and let $M'_{i,j}$ denote the $(i, j)^{\text{th}}$ minor of M (i.e., the submatrix of M obtained by deleting the i^{th} row and j^{th} column). Since

$$\begin{aligned} \nu(\det(M)) &\leq \sum_{j=1}^n \nu(M_{1,j}) \nu(\det(M'_{1,j})) \\ &\quad \text{(by Laplace expansion formula for determinants)} \\ &\leq \sum_{j=1}^n \left(\nu(M_{1,j}) \prod_{i=2}^n \nu(M_i) \right) \\ &\quad \text{(since every row of } M'_{1,j} \text{ is a subrow of some } M_i, i \geq 2) \\ &= \prod_{i=2}^n \nu(M_i) \left(\sum_{j=1}^n \nu(M_{1,j}) \right) \\ &\leq \prod_{i=1}^n \nu(M_i). \end{aligned}$$

Problem 8.13

Let $\delta = \alpha - \beta$. Then $\langle \delta, \beta \rangle$ is a zero of $\tilde{f}(x, y) = f(x + y)$ and β is a zero of $g(y)$. Thus δ is a nonzero root of the polynomial

$$R(y) = \text{Resultant}_y(\tilde{f}(x, y), g(y)),$$

and

$$\Delta = |\delta| > \frac{1}{1 + \|R\|_\infty}.$$

Now,

$$\|R\|_\infty \leq \|R\|_1 \leq \|\tilde{f}(x, y)\|_1^n \|g(y)\|_1^m.$$

Observe that the 1-norm of $\tilde{f}(x, y)$ is taken over $\mathbb{Z}[\curvearrowright, \curvearrowleft]$. Now writing

$$f(x) = a^m x^m + a_{m-1} x^{m-1} + \cdots + a_0$$

we have

$$\begin{aligned} \tilde{f}(x, y) &= f(x + y) = a^m (x + y)^m + a_{m-1} (x + y)^{m-1} + \cdots + a_0 \\ &= \sum_{i=0}^m a_i \sum_{j=0}^i \binom{i}{j} x^{i-j} y^j \\ &= \sum_{j=0}^m \left(\sum_{i=j}^m a_i \binom{i}{j} x^{i-j} \right) y^j \\ &= \sum_{j=0}^m A_j(x) y^j. \end{aligned}$$

Thus

$$\begin{aligned} \|\tilde{f}(x, y)\|_1 &= \|A_m(x)\|_1 + \cdots + \|A_0(x)\|_1 \\ &\leq \sum_{j=0}^m \|f\|_\infty \sum_{i=j}^m \binom{i}{j} \\ &\leq \|f\|_\infty \sum_{i=0}^m \sum_{j=0}^i \binom{i}{j} \\ &\leq \|f\|_1 \sum_{i=0}^m 2^i < \|f\|_1 2^{m+1}. \end{aligned}$$

Thus

$$\|R\|_\infty < 2^{n(m+1)} \|f\|_1^n \|g\|_1^m,$$

and

$$\Delta > \frac{1}{1 + \|R\|_\infty} > 2^{-(n+1)(m+1)} \|f\|_1^{-n} \|g\|_1^{-m}.$$

Problem 8.15

(i) Let the roots of f be enumerated as $\xi_1, \xi_2, \dots, \xi_n$. Now consider

the determinant of the following Vandermonde matrix:

$$V_n = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \xi_1 & \xi_2 & \cdots & \xi_n \\ \vdots & \vdots & \ddots & \vdots \\ \xi_1^{n-2} & \xi_2^{n-2} & \cdots & \xi_n^{n-2} \\ \xi_1^{n-1} & \xi_2^{n-1} & \cdots & \xi_n^{n-1} \end{bmatrix}.$$

Since $\det(V_n V_n^T) = \text{Discriminant}(f) \neq 0$ (f is square-free) and since f is integral, we have

$$|\det V_n| \geq 1.$$

Now if we subtract the j^{th} column of V_n from its i^{th} column, then this elementary matrix operation does not change $\det V_n$, and we have

$$\det V_n = \begin{vmatrix} 1 & 1 & \cdots & 0 & \cdots & 1 \\ \xi_1 & \xi_2 & \cdots & \xi_i - \xi_j & \cdots & \xi_n \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \xi_1^d & \xi_2^d & \cdots & \xi_i^d - \xi_j^d & \cdots & \xi_n^d \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \xi_1^{n-2} & \xi_2^{n-2} & \cdots & \xi_i^{n-2} - \xi_j^{n-2} & \cdots & \xi_n^{n-2} \\ \xi_1^{n-1} & \xi_2^{n-1} & \cdots & \xi_i^{n-1} - \xi_j^{n-1} & \cdots & \xi_n^{n-1} \end{vmatrix}.$$

Using the Hadamard inequality, we have

$$\begin{aligned} 1 &\leq |\det V_n| \\ &\leq \left(\sum_{d=1}^{n-1} |\xi_i^d - \xi_j^d|^2 \right)^{1/2} \prod_{k \neq i} \left(\sum_{d=0}^{n-1} |\xi_k|^{2d} \right)^{1/2}. \end{aligned}$$

Now

$$\begin{aligned} |\xi_i^d - \xi_j^d| &= |\xi_i - \xi_j| |\xi_i^{d-1} + \xi_i^{d-2}\xi_j + \cdots + \xi_j^{d-1}| \\ &\leq d|\xi_i - \xi_j| |\xi_i|^{d-1}, \end{aligned}$$

assuming $|\xi_i| \geq |\xi_j|$.

Hence,

$$\sum_{d=1}^{n-1} |\xi_i^d - \xi_j^d|^2 < \left(\frac{n^3}{3} \right) |\xi_i - \xi_j|^2 \max(1, |\xi_i|)^{2n-2}.$$

Similarly,

$$\sum_{d=0}^{n-1} |\xi_k|^{2d} < n \max(1, |\xi_k|)^{2n-2}.$$

Thus, combining these inequalities, we get

$$1 < |\xi_i - \xi_j| \left(n^{n-1} \left(\frac{n^3}{3} \right) \right)^{1/2} \prod_{k=1}^n \max(1, |\xi_k|)^{n-1}.$$

and

$$|\xi_i - \xi_j| > \frac{\sqrt{3}}{n^{(n+2)/2} \mathcal{M}(f)^{n-1}}.$$

Thus since $\mathcal{M}(f) \leq \|f\|_1$, we have the result

$$\Delta(f) > \frac{\sqrt{3}}{n^{(n+2)/2} \|f\|_1^{n-1}}.$$

(ii) Assume that $f(x)$ is not square-free. Then its square-free factor $g(x)$ is given by

$$g(x) = \frac{f(x)}{\text{GCD}(f(x), f'(x))}, \quad \text{and } \|g\|_1 \leq 2^n \|f\|_1.$$

Then writing $m = \deg(g)$,

$$\begin{aligned} \Delta(f) &= \Delta(g) \\ &> \frac{\sqrt{3}}{m^{(m+2)/2} \|g\|_1^{m-1}} \\ &> \frac{\sqrt{3}}{m^{(m+2)/2} (2^n \|f\|_1)^{m-1}} \\ &> \frac{\sqrt{3}}{2^{n(n-1)} n^{(n+2)/2} \|f\|_1^{n-1}}. \end{aligned}$$

Problem 8.18

Let $S \subseteq \mathbb{R}^\times$ be a semialgebraic set. Let $p \in S$ be an arbitrary point of S and consider a small open neighborhood of p defined as follows:

$$N_{S,\epsilon}(p) = S \cap \{q : \|p - q\|_2 < \epsilon\},$$

for some $\epsilon > 0$. Clearly, $N_{S,\epsilon}$ is a semialgebraic set defined by a set of polynomials $\mathcal{F}_\epsilon$. Now consider a cylindrical algebraic decomposition $\mathcal{K}_\epsilon$ of $\mathbb{R}^\times$ defined by $\mathcal{F}_\epsilon$. Note that $\mathcal{K}_\epsilon$ has finitely many cells and that for every ϵ there is an open connected cell $C_\epsilon \subseteq N_{S,\epsilon}(p)$ such that

$$p \in C_\epsilon \subseteq S.$$

Thus S is locally connected.

Problem 8.21

The proof is in two steps: First, we shall construct a sequence of complex quantified sentences Φ_k 's each equivalent to a single polynomial as follows:

$$\Phi_k(z_0, z_k) : z_k^{2^{2^k}} = z_0.$$

In the second step, we shall obtain a sequence of real quantified sentences Ψ_k 's each being an equivalent real version of Φ_k .

$$\begin{aligned} \Psi_k(\Re(z_0), \Im(z_0), \Re(z_k), \Im(z_k)) : \\ \left(\Re \left([\Re(z_k) + i\Im(z_k)]^{2^{2^k}} \right) = \Re(z_0) \right) \\ \wedge \left(\Im \left([\Re(z_k) + i\Im(z_k)]^{2^{2^k}} \right) = \Im(z_0) \right), \end{aligned}$$

where $\Re(z)$ and $\Im(z)$ stand, respectively, for the real and imaginary parts of z . Now renaming $x = \Re(z_k)$ and $y = \Im(z_k)$ and letting $z_0 = 1$ (i.e., $\Re(z_0) = 1$ and $\Im(z_0) = 0$), we get the final Tarski sentence.

BASE CASE: $k = 0$, $N = 2^{2^0} = 2$:

$$\Phi_0(z_0, z_1) : z_1^2 = z_0.$$

Thus Ψ_0 is

$$\begin{aligned} \Psi_0(\Re(z_0), \Im(z_0), \Re(z_1), \Im(z_1)) : \\ (\Re(z_1)^2 - \Im(z_1)^2 = \Re(z_0)) \wedge (2\Re(z_1)\Im(z_1) = \Im(z_0)), \end{aligned}$$

or, after renaming

$$\Psi_0(x, y) \equiv \Psi_0(1, 0, x, y) \equiv (x^2 - y^2 - 1 = 0) \wedge (xy = 0).$$

INDUCTION CASE: $k > 0$, $N = 2^{2^k}$:

We would like to define

$$\Phi_k(z_0, z_k) \equiv (\exists w) \left[(\Phi_{k-1}(w, z_k)) \wedge (\Phi_{k-1}(z_0, w)) \right].$$

Thus Φ_k is equivalent to saying

$$\left(z_k^{2^{2^{k-1}}} = w \right) \wedge \left(w^{2^{2^{k-1}}} = z_0 \right) \equiv z_k^{2^{2^k}} = z_0.$$

However, in this case the formula size becomes “exponentially large”! We avoid the exponential growth by making sure that only one copy of Φ_{k-1}

appears in the definition of Φ_k as follows:

$$\begin{aligned}\Phi_k(z_0, z_k) &\equiv (\exists w) (\forall z', z'') \\ &\quad \left[[(z' = z_k) \wedge (z'' = w)] \vee [(z' = w) \wedge (z'' = z_0)] \right. \\ &\quad \left. \Rightarrow (\Phi_{k-1}(z', z'')) \right].\end{aligned}$$

Thus

$$\begin{aligned}\Psi_k(\Re(z_0), \Im(z_0), \Re(z_k), \Im(z_k)) \\ \equiv (\exists u, v) (\forall x', y', x'', y'') \\ \left[[(x' \neq \Re(z_k)) \vee (y' \neq \Im(z_k)) \vee (x'' \neq u) \vee (y'' \neq v)] \right. \\ \wedge [(x' \neq u) \vee (y' \neq v) \vee (x'' \neq \Re(z_0)) \vee (y'' \neq \Im(z_0))] \\ \left. \vee \Psi_{k-1}(x', y', x'', y'') \right].\end{aligned}$$

Finally, after renaming, we have

$$\Psi_k(x, y) \equiv \Psi_k(1, 0, x, y).$$

Bibliographic Notes

The study of *real closed field* was initiated by Artin and Schreier in the late twenties in order to understand certain *nonalgebraic properties* (such as positive-ness, reality, etc.) of the numbers in an algebraic number field. (See [8].) These topics are also discussed extensively in the standard algebra textbooks. See, for instance: Jacobson [105] and van der Waerden [204].

Fourier is usually credited with the idea of solving an algebraic equation (for a real solution) in two steps: a *root isolation* step followed by an *approximation* step. These ideas lead to what has come to be known *Sturm's theory*. During the period between 1796 and 1820, Budan (1807) and Fourier (1796 and 1820) achieved initial success in devising an algorithm to determine an *upper bound* on the number of real roots of a real polynomial in any given interval. (Also, see Problem 8.6.) In 1835, Sturm [195], building on his own earlier work, finally presented an algorithm that determines the *exact* number of real roots of a real polynomial in a given interval. Some later generalizations are credited to Tarski.

Our discussion of the *Sturm-Tarski theorem* is based on the exposition given in Mishra and Pedersen [148]. The formulation of Corollary 8.4.6 is due to Ben-Or et al. [19]. (Also, see Mishra and Pedersen [148].) Our discussion of the bounds on the roots and root separation are based on the papers by Collins and

Horowitz [54], Mignotte [144], and Rump [180]. Also, consult the classic book of Marden on the “geometry of the zeros of a polynomial” [139].

For a discussion of the representation and applications of real algebraic numbers, consult the papers by Loos [133] and Coste and Roy [55]. Our complexity analysis of various operations on the real algebraic numbers uses fast implementations of the Sturm sequence and resultant computation, first given by Schwartz and Sharir [185].

A generalization of Sturm’s theorem to the multivariate case, embodying a decision procedure for the *first-order theory of real closed fields*, was devised by Tarski [200] in the early thirties, but published in 1948, after the Second World War. However, the original algorithm of Tarski has a nonelementary complexity and the computational infeasibility of the algorithm made it only of theoretical interest. In 1973, Collins [50] discovered the algorithm, discussed here, and demonstrated that it has a polynomial complexity, if the number of variables is kept constant. Many of the key ideas in Collins’ work had, however, appeared in the work of Koopman and Brown [119]. Because of its wide range of applicability, Collins’ work has been surveyed for various groups of readership, as in the papers by Arnon [5], Davenport [56], Hong [100], Pedersen [161] and Schwartz and Sharir [185]. For an extended bibliography in the area, see Arnon [6].

In the last five years or so, certain significant improvements have been achieved in the time complexity (both sequential and parallel) of the decision problem for first-order theory of real closed fields. Some of the influential works in this area are due to: Ben-Or et al. [19], Canny [40, 41], Davenport and Heintz [57], Fitchas et al. [72], Grigor’ev [85], Grigor’ev and Vorobjov [86], Heintz et al. [92], Renegar [166-169] and Weispfenning [208].

Assume that the given Tarski sentence involves the multivariate polynomials $\mathcal{F}$ in n variables and has ω alternating quantifiers

$$\mathcal{Q}_1, \dots, \mathcal{Q}_\omega, \quad \mathcal{Q}_i \neq \mathcal{Q}_{i+1},$$

with the i^{th} quantifier involving n_i variables.

The time complexity of the currently best sequential algorithm is

$$\left(|\mathcal{F}| \deg(\mathcal{F}) \right)^{\prod O(n_i)},$$

and the time complexity of the currently best parallel algorithm is

$$\left[\prod O(n_i) \left(|\mathcal{F}| \deg(\mathcal{F}) \right) \right]^{O(1)}.$$

For more details, see the survey paper by Renegar [170].

On the general subject of *real algebra*, the reader may consult the following books by Bochnak et al. [21] and Benedetti and Risler [20]. Also, consult the special issue of *J. Symbolic Computation* entitled “Algorithms in Real Algebraic Geometry” (Volume 5, Nos. 1 & 2, February/April 1988).

Problem 8.8 is based on the work of Ben-Or et al. [19]. Problem 8.9 is taken from Rump’s paper (lemma 2) [180]. Problem 8.10 is due to Mignotte [144]. The inequality of Problem 8.11 was derived by Collins and Horowitz [54]. Problem 8.13 is motivated by [54], though Collins and Horowitz’s techniques

will produce a somewhat sharper bound. Problems 8.14 and 8.15 are due to Mahler [136, 137]. Problem 8.16 (ii) is from [55]. A solution to *robot motion planning problem* (Problem 8.20) was given by Reif [166] and Schwartz and Sharir [185]. A more efficient solution is due to Canny [40] and uses a stratification to study connectivity of semialgebraic sets. Problems 8.21, 8.22 and 8.23 are due to Davenport and Heintz [57].

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